1. End User Manual 2
2. CPO DC Meter Manual 5
3. Concept of Use for DC Meter Tariffs 8
4. Techniques & Notes for Developers 10
5. Trusted Screen Overlay (TSO) 12
    5.1 Display Format 18
    5.2 FPGA IP core 33
        5.2.1 Pixel interceptor to microblaze interface 38
    5.3 Pairing TSO and DCM 39
        5.3.1 Trust Anchor Communication 40
    5.4 Rendering of overlay screen 42
6. DC Meter Approved Firmware Version 45
7. Charger power derate on TCU 46

End User Manual

General Information

For Electric Vehicle Supply Equipment (EVSE) in some markets, laws exist which require the regulation of billing on public charging facilities.

This manual outlines how a user can check their bill inline with these requirements.

During charging a certified DC meter will generate a digital receipt with a cryptographic signature to ensure it has come from a trusted source. This signature can then be checked with the certified transparency software. Once verified, it is ensured that all information in the receipt was generated by the certified DC meter and has not been tampered with.

The interactions with metering and the user are mostly between the Charge Point Operator (CPO) and the user. For support please contact your CPO.

Transparency Software

Transparency software and instructions for use can be obtained at:

https://www.safe-ev.de/en/transparency_software.php

Activities at the charger

To successfully check the billing receipt you must do the following steps at the charger:

1. Start a charge on your vehicle as usual
    a. Follow the prompts on the charger's LCD screen to begin the charging session.
    b. The DC meter will automatically detect the start of charge based on measured power output to the vehicle. There is no need for any further user input.
2. Either during or after the charging session, take note of the 'Session Signature' displayed on the legally relevant display
    a. It is recommended that a photo is taken to guarantee that the information is correctly recorded.
    b. The three screens shown below will cycle continuously, displaying for 5 seconds each.







Verifying the bill

To verify the billing receipt you must:

- Obtain the billing receipt from the CPO
    - The method of receiving this will vary depending on the provider
- Download and read the instructions of the transparency software
- Check that the 'Session Signature' included in the bill is the same as noted at the charging station
- Use the transparency software to load the billing receipt and verify the data is trusted
- Once the data is confirmed as trusted, all information in the SAFE billing receipt can be used to verify the CPO pricing model
    - 'Energy Delivered' and 'Duration' of charge session is typically used to calculate pricing by the CPO
    - Pricing models will vary depending on the provider

Display Icons

Below is the explanation of the icons on the legally relevant display from the metering device to assist in extracting the relevant information.

Icon	Purpose	Explanation
	Public Key Fingerprint	The public key of the metering device. Can be used to verify the authenticity of the digital receipt
	Play - Charge	Indicates the information about a charge session's start/stop times
	Record - Charge	Indicates the information recorded during a charge session
	Tariff Name	The name of the tariff applied (this can be blank)
	Session Signature (Receipt)	Unique ID of the charging session. Used to link to digital receipt provided by CPO
	Firmware Hash	Can be used to verify the identify of firmware version
	Present Time	Present local time on the metering device
	Start Time	Local time when charge session started
	Stop Time	Local when charge session ended
	Duration	Duration of a charge session

	Energy Delivered	Cumulative total of energy delivered in a session
	Firmware Version	Version of firmware the metering device is running
	Error: No communication	Screen has lost connection with metering device. Note: Will appear on unit when booting up. When on this screen the firmware version displayed is the version of the Trusted Screen Overlay
	Error: Out Of Service	Metering device is out of service

CPO DC Meter Manual

This page defines the interactions that the CPO can have with the DC Meter.

- Requirements
- SAFE Initiative Data
    - OCMF Compatibility
    - Transport Mechanism
        - has.to.be DataTransfer message for initial communication
        - Tritium-specific DataTransfer based protocol for SAFE Initiative Data
- Tariffs
    - Tritium-specific OCPP configuration setting for default Tariff information
    - Tritium-specific OCPP read-only configuration setting for active Tariff information

Requirements

The Charge Point Operator (CPO) must:

- Register DC Meter's public key with German government
- Provide the SAFE data receipt to end user as received from the DC Meter (when requested by user)
- Communicate the calculation of the tariff to the End User.

SAFE Initiative Data

The billing data determined by the DC Meter will be passed to the CPO as a file using the OCMF format to support the SAFE initiative.

This will be passed through OCPP, initially via a Tritium-specific DataTransfer message, and eventually via an OCA-approved enhancement to the main OCPP specification.

CPO must check the signature of received data to use for billing.

OCMF Compatibility

The DC Meter complies with the OCMF v1.0 (Open Charge Metering Format) which can be obtained here:

http://hers.abl.de/SAFE/Datenformat_OCMF/Datenformat_OCMF_v1.0.pdf

To support the flexibility of external billing arrangements, reading entries are created every 15 minutes of charging with energy reading value. The end of transaction reading entry will contain the total session energy usage.

Tritium's DC Meter makes use of two optional manufacturer extension fields:

- VS - in user assignment portion of main data - contains a "Session Signature" that is globally unique to the charge session.
- UT - contains the name of the tariff active in this charge (max 16 characters)

Transport Mechanism

The DC Meter signed SAFE data allows it to be transformed and go through intermediaries whilst still ensuring its authenticity and integrity. The DC Meter will send its SAFE message in base64 to the charger over MQTT which will then pass this message to the CSMS.

has.to.be DataTransfer message for initial communication

https://has-to-be.com/en/development-resources/transparenzsoftware/ocpp-integration/ covers sending initial info to has.to.be on startup

Tritium will add a new type of `SIGNED_OCMF`, that uses the DataTransfer below rather than sending individually signed meter values.

Tritium-specific DataTransfer based protocol for SAFE Initiative Data

As certified meters are required to independently detect the start and stop of energy transfer periods, there may be multiple meter sessions associated with each OCPP transaction.

Tritium will collect all the sessions associated with a transaction on the charger, and then send them to the CSMS as part of a single message at the end of the transaction.

Message from charger to CSMS following completion of charging session:

```
[2, "12341234", "DataTransfer",
  {
    "vendorId": "Tritium",
    "messageId": "CalibratedMeterSessions",
    "data": "{\"transactionId\":\"12345\",\"sessions\":[\"<signed OCMF payload for first energy transfer
period>\",\"<signed OCMF payload for second energy transfer period>\"]}"
  }
]
```

Payload from example DataTransfer message above (i.e. after removing the escape characters and decoding as JSON):

```
{
    "transactionId": "12345",
    "sessions": [
        "<signed OCMF payload for first energy transfer period>",
        "<signed OCMF payload for second energy transfer period>"
    ]
}
```

The CSMS will be expected to respond with a regular OCPP DataTransfer confirmation message with the status "Accepted" and no data payload.

Tariffs

The DC Meter supports a tariff name to be displayed locally on the Trusted Screen Overlay and logged with the charge session information (as the OCMF optional "UT" field in the first record in the signed transaction data). No fiscal calculations are supported.

The DC Meter is configured in production with a default tariff name that will be used on power up.

The CPO can choose to update the active tariff throughout the meter's life through OCPP. Alternatively the Tariffs can be left with the default name.

Tariff name changes are signed with the CPO's private key. The signature is verified with the CPO public key which is pre-written into the meter's memory also during production. The CPO public key will conform to SECP256R1 (Key Curve), ASN.1 DER (Format), RFC-5480 (Standard).

Once tariffs are sent to the meter they will be used until new tariffs are sent or a power cycle. The tariff for a given charge session will be set once power delivery has started and changing the active tariff after this time will only apply to the next charge session.

Tritium-specific OCPP configuration setting for default Tariff information

Whenever power is cycled, the certified DC Meter will always revert to its factory default tariff setting. To update the active tariff, Tritium offer a custom OCPP configuration setting, *"DefaultCalibratedMeterTariff"* that allows a CPO to provide a signed tariff name in the format "<TARIFF NAME>;<TARIFF SIGNATURE>".

The tariff name consists up to 16 alphanumeric ASCII characters. If less than 16 characters long it will be padded for display with leading or trailing spaces as appropriate. To be accepted as a valid tariff by the DC Meter, the tariff name must be followed by a semi-colon, and then a hexadecimal signature generated using the CPOs private key.

If this setting is empty, then the DC Meter will continue to use its existing tariff settings, and will revert to the factory default tariff setting if power cycled. There is not presently any specific mechanism provided to request that the meter revert to the factory default tariff setting.

If this setting is not empty, then its value will be sent to the DC Meter as the active tariff after the DC Meter first reports its serial number and public key to the OCPP client. The DC Meter will only accept the update if the given signature matches the given tariff name.

Tariff changes are not applied immediately if a charging session is in progress. In these cases, the tariff change will be queued internally and applied once the energy delivery session is completed.

Tritium-specific OCPP read-only configuration setting for active Tariff information

Changes to the default tariff may be rejected if the signature was created using a private key that does not match the public key installed in the meter. To assist in debugging these cases, additional read-only settings are published by the Tritium OCPP client:

- *HUx.DCMActiveTariff*: The tariff currently active in the DC Meter for a given user unit
- *HUx.DCMQueuedTariff*: Differs from the active tariff only when a tariff change is queued to become active at the end of the current energy delivery session
- *HUx.DCMCPOPublicKey*: The factory-installed public key that the DC Meter expects to be used to sign tariff changes

These settings will only be published if a user unit has a DC Meter installed.

Concept of Use for DC Meter Tariffs

CPO DC Meter Manual covers how to set the active tariff name via OCPP, but doesn't explain how Tritium expect tariff names to be used.

This supporting document covers those expectations.

Core Operating Concept

Tritium's core operating concept for DC Meter tariffs is that the role of the tariff name is to capture any information that is not otherwise stored as part of the signed OCMF records generated by the DC meter.

Tritium also expect that the tariff name may also capture information that could technically be inferred from the signed OCMF record, but only with difficulty.

The communication of the meaning of tariff names will take place external to the charger (e.g. on the CPOs website, or on a local information sticker). To allow for tariff changes without affecting reconciliation of historical transactions, it is also expected that some kind of version number would be included in the tariff names (at least for the second and subsequent versions of the tariff definition).

Note: not all tariff schemes described in this document are expected to be permitted in all regulatory environments. The use cases cover how to manage Tritium chargers in the event that the local regulatory environment *does* allow for the described tariff model, and a network operator has chosen to use it.

Case 1: Flat Tariff

- Example tariff names: *TRI_FLAT_1*, *TRI_FREE_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP or else via the factory default DC Meter setting.

This is the simplest possible case: the tariff name simply refers to a particular price per kilowatt hour. The signed OCMF records include the amount of energy delivered, allowing this to be calculated in a straightforward manner.

Free energy delivery is simply the special case where the price per kWh is zero.

Case 2: Time of Day Based Tariff Variation

- Example tariff names: *TRI_AEST_1*, *TRI_CET_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP.

When tariffs vary by time of day, they vary based on the *local* time where the charger is located, while the signed OCMF records will capture session times in UTC.

In theory, the time zone for the charging session can be inferred from the public key of the DC Meter (since the installation location of the meter must be registered with the appropriate authorities).

In practice, it may be clearer to encode the local time zone information as part of the tariff name.

The signed OCMF records published by the DC Meter capture the energy delivered in 5 minute increments for the first 6 hours, and then in 1 hour increments after that. Tariff definitions will need to cover how increments that span a tariff rate change are handled, with Tritium's expectation being that the most common practice will be to assume constant power delivery across the time interval, and charge for energy accordingly. However, some regulatory environments may require that the entire interval be charged at whichever tariff is lower. Either way, that information is expected to be part of the tariff definition, rather than requiring dynamic changes to the active tariff name.

Case 3: Average Power Based Tariff Variation

- Example tariff names: *TRI_SESSION_1*, *TRI_INTERVAL_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP or else via the factory default DC Meter setting.

As the signed OCMF records include both energy delivery and duration information, they can be used to calculate the average charging power for both the entire session and for each recorded interval within that session.

This means that billing based on the average power achieved can be handled as part of the tariff definition, making this case equivalent to a flat tariff from an operational perspective.

Case 4: Maximum Power Based Tariff Variation (external billing discounts permitted by regulators)

- Example tariff name: *TRI_PWR_LIMIT_1*
- Supported in pk.7.0.0 by setting *DefaultCalibratedMeterTariff* via OCPP or else via the factory default DC Meter setting.

Regulatory environments are expected to differ in whether or not they allow network operators to apply additional external surcharges and discounts beyond the base tariff based on energy delivered (for example, consider fuel discount vouchers and credit card surcharges that affect the actual price paid by drivers for fuel, but don't affect the nominal price-per-litre of that fuel).

In regulatory environments where at least external discounts are permitted, then maximum power based tariffs can be handled as flat tariffs at the highest rate, combined with a percentage or per kWh discount for accepting a lower power limit on the charging session.

For example, a network operator might charge $1.29 per kWh for a session at the full power a HPC station can deliver, but only $0.79 per kWh if the driver agrees to explicitly cap their session at a maximum charge rate of 50 kW. If acceptable to regulators, this scenario could be modeled as a flat tariff of $1.29 per kWh, with an optional external discount of $0.50 per kWh for accepting the lower power limit.

With this approach, while the signed OCMF data could be used to demonstrate that a particular session had gone above 50 kW for one or more measurement periods, it wouldn't be able to conclusively distinguish sessions that were hard limited to 50 kW on the EVSE side from those where unlimited power was permitted, but the EV simply never requested more than 50 kW.

Case 5: Maximum Power Based Tariff Variation (external billing discounts NOT permitted by regulators)

- Example tariff names: *TRI_HIGH_PWR_1*, *TRI_LOW_PWR_1*
- Not supported in pk.7.0.0
- Expected to be supported in pk.8.0.0 via the *SetDynamicPricing* transaction-specific *DataTransfer* message

If external billing discounts are not permitted in a particular regulatory environment, then dynamic agreements to a different charging rate will presumably need to be reflected in the tariff name set for that charging session. While not expected to be part of the initial DC Meter compatible OCPP client release, Tritium's intention is to support this use case through the *SetDynamicPricing* transaction-specific *DataTransfer* message.

The key operational challenge with this approach is that there's only a limited time window between the Charging Station Management System (CSMS) receiving a *StartTransaction* message indicating that a link has been established to the EV and a charge session successfully authorized, and the DC Meter locking in the active tariff due to the start of energy delivery to the vehicle. While that time window is several seconds long, the failure mode where the *SetDynamicPricing* message arrives too late still needs to be taken into account.

Consider the same example as in case 4, where a network operator charges $1.29 per kWh for a session at the full power a HPC station can deliver, but only $0.79 per kWh if the driver agrees to explicitly cap their session at a maximum charge rate of 50 kW. This scenario can be modeled as two distinct tariffs, one for the low power rate, and one for the high power rate. To ensure customers aren't overcharged for low power sessions, the low power tariff and smart charging profile can be set as default, and then *RemoteStart/SetChargingProfile* and *SetDynamicPricing* used to enable the higher rate when appropriate. If *SetDynamicPricing* is sent second, then an untimely loss of connectivity may lead to a user being undercharged, but it won't lead to them being overcharged.

With this approach, the signed OCMF data explicitly indicates which tariff was in effect at the time of the charger session, and this same tariff name would be displayed on the DC Meter trusted screen overlay while the charge session was in progress.

Case 6: Dynamic Free Mode Tariff Variation

- Example tariff name: *TRI_OFFLINE_1*
- Indicated in pk.7.0.0 by an OCPP transaction ID of "-1" appearing in the signed OCMF records for at least part of the session
- Potentially better supported in pk.8.0.0 by offering an *OfflineCalibratedMeterTariff* setting via OCPP that is used in preference to the default tariff when the OCPP link is down

In the initial implementation, the fact a charge session took place while the charger was offline needs to be inferred from the fact that there's no real OCPP transaction ID set in the signed OCMF records.

A new OCPP setting would allow a specific tariff name to be sent to the DC Meter when the charger determined that the OCPP link is down.

Techniques & Notes for Developers

- 1 Pairing of TSOs & DC Meters
  - 1.1 Trust Root System & Mock Trust Root
  - 1.2 Checking if TSO is Paired/Programmed
  - 1.3 Checking if a DC Meter is Paired/Programmed
- 2 Resetting the JTAG lockout

Pairing of TSOs & DC Meters

Trust Root System & Mock Trust Root

During IQC, there is a step that pre-loads a TSO and DC Meter with cryptographic information that allows them to communicate & verify their "trust" in each other.
The cryptographic pairing process can only be performed by the air-gapped trust root system (which is located in Amsterdam).
If a DC Meter is not paired to the installed TSO, then it will fail to connect (showing error TSO_NO_SESSION).

It is possible to upload firmware to a TSO and DC Meter with a **mock trust root**.
This will bypass the cryptographic trust validation, and can only be used in testing and development.
To create the firmware image for a mock-trusting meter or TSO, the following build options must be set:

DC Meter:

DCMCONFIG_MOCK_TRUST_ANCHOR=1
TSOCLIENT_FULL_SESSION_COUNTER_LIMIT=1

TSO:

MOCK_TRUST_ANCHOR=1

These build options can be modified within the CCS project using the "Project > Properties > Predefined Symbols" window.

Checking if TSO is Paired/Programmed

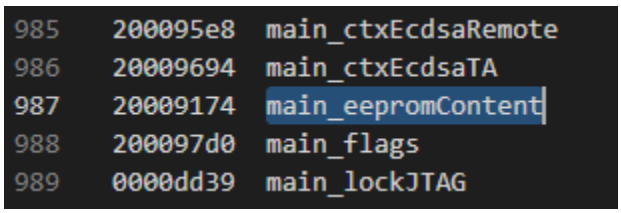
If there is no sticker on the TSO or DC Meter, it is impossible to tell visually if they have been paired or not.
Once the state of a TSO is known, the serial number of its paired DC Meter should be marked on the TSO sticker.
To check if a TSO is programmed, build the PK or ISD TSO release images from CCS, then start a debug session with "Load Symbols Only" enabled.

Once connected to the target, using the memory browser and check the following location in memory:

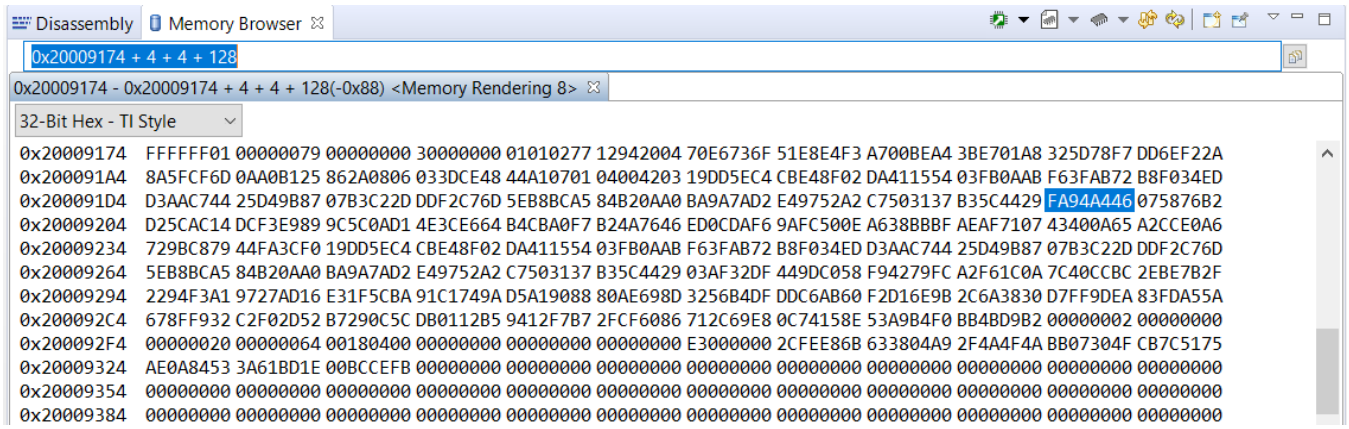
<root> + 4 + 4 + 128

The variable **<root>** is either 0x20009174 for ISD and 0x2000667c for PK at the time of writing.

The exact value of **<root>** can be confirmed by searching the *.map file from the build output for "main_eepromContent".
The address in the left column should show where the variable is mapped in static storage, as seen below:



If the memory looks like the following image, then your TSO is programmed. The next 64 bytes show the public key that was programmed into the EEPROM.
Note that you cannot access the EEPROM directly from CCS, because it's abstracted behind a memory controlled (i.e. not mapped into the CPU memory map).



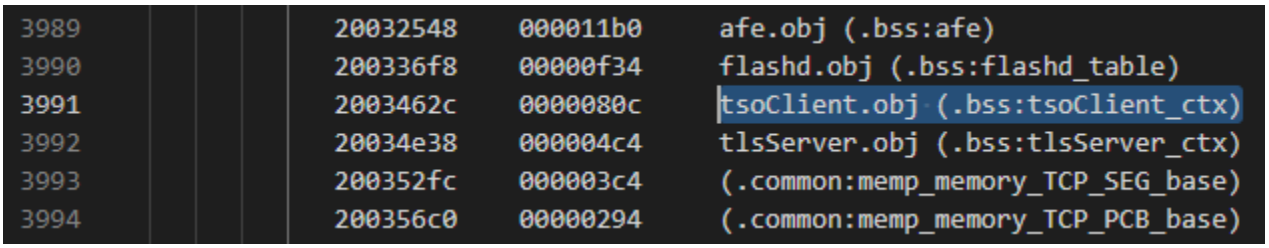
If the memory location contains "FFFFFFFF FFFFFFFF ...", then the EEPROM has not been programmed and the TSO will not work with any DC Meter.

Checking if a DC Meter is Paired/Programmed

The same technique as above is used - connect to the DC meter and use the memory browser to search for this location in memory:

<tsoClient_ctx> + 0x69B

The value of <tsoClient_ctx> can be found in the *.map file of the build by searching for "tsoClient_ctx".
See the below screenshot for an example from firmware version 1.012:

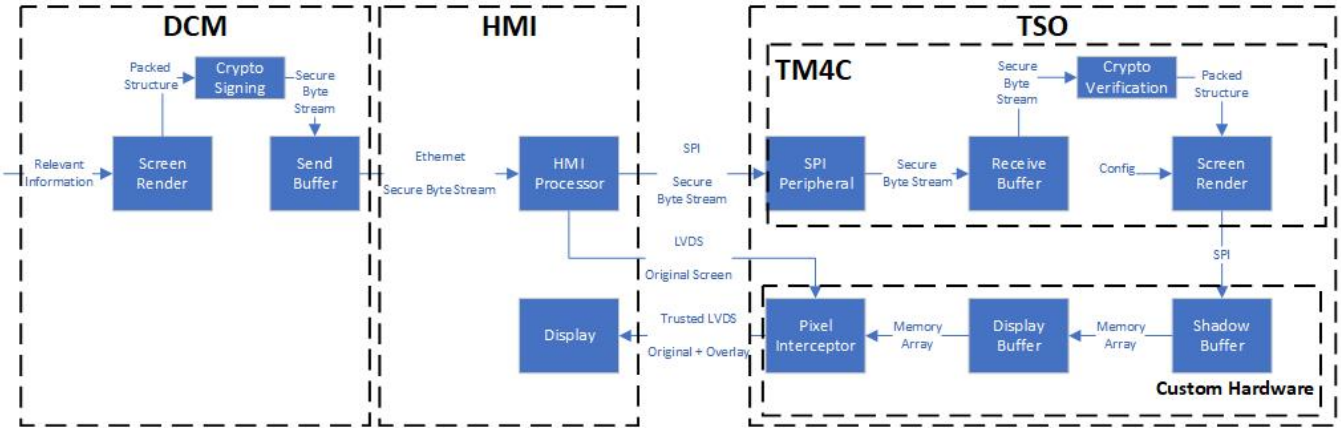


In the case shown above, the root memory address is **0x2003462C**, and 0x80C bytes have been reserved for it.
Note that you will need to allow the DC Meter to run with an active network connection before the public key is filled in.

Resetting the JTAG lockout

Refer to this wiki page: Unlocking JTAG port on TM4C device

Trusted Screen Overlay (TSO)



The TSO receives secure information from the DC meter and forces the legally relevant information onto the screen regardless of any other device /software.

The overlay of the TSO is clearly differentiable from the rest of the display.

Related pages

- Display Format
- FPGA IP core
  - Pixel interceptor to microblaze interface
- Pairing TSO and DCM
  - Trust Anchor Communication
- Rendering of overlay screen

Configuration

See DC Meter Configuration.

Firmware Version

The TSO firmware is programmed in the factory and is never changed. The firmware version is displayed on the screen whenever not connected to the DC Meter including during start-up. The firmware version covers both the TSO's TM4C and the FPGA/microblaze (aka Custom Hardware) and is manually controlled. The version will be incremented if there is any change to the source code. The version will be in a format of:

V[Major].[Minor].[Build]_[Variant] - e.g. V01.00.01_ISD

- Major version is incremented if there is a major change to functionality.
- Minor version is incremented if there is a minor change to functionality, or to reflect many build version level changes.
- Build version is incremented for any change to source code, normally used for insignificant changes/bug fixes.
- Variant represents the build configuration. There can be several build configurations for the same version. Current allowed variants are:
  - PK
  - ISD

The TSO build variant has a corresponding configuration parameter inside the DC Meter (See DC Meter Configuration, section "TSO Configuration"). If the DC Meter configuration is not matched with the firmware variant on the TSO, then the TSO will not display the legally relevant data and instead display a "No Connection" screen with the firmware version printed. The DC Meter will also go to "Out of Service".

Preventing Update

The TSO has several mechanisms to prevent changing firmware.

- TSO is installed in the charger behind a tamper evident seal
- TSO's TM4C
    - Firmware disables the programming header once key is generated and signed by trust root
    - No bootloader present in firmware
- TSO's FPGA
    - Device program is encrypted with security key, FPGA has encrypted key loaded into its memory. Can only be programmed with that key
    - Only communication interface with TM4C which will not bootload the device

Communications Between TSO and DC Meter Controller

Security

Pairing process

See Pairing TSO and DCM.

TSO requires the session to be initiated by the DC meter, otherwise the displayed message cannot be changed.

If TSO does not receive any frame with valid signature for 5 seconds or signature is incorrect it will go to disconnected mode.

Session Sequence

1. DCM will exchange its public key and digital signature provided from the Trust Root to the TSO. The DCM will poll the status of the TSO, which includes the TSO public key and digital signature. Each device will independently verify that their received key can be verified and trusted.
2. DCM will stop an active session (if any). After stopping a session, each device will reset their replay protection counters.
3. DCM will receive a random 32-bit session ID from the TSO. This session ID is to be used in each communication frame, until the session is stopped.
4. DCM will enter an active session with the TSO. The DCM will generate a 32-bit random number and send it to the TSO. The TSO will clear any previous errors and reply with the same random number supplied by the DCM. The DCM will confirm that the number is the same and continue the session.
5. DCM will load the Character Map into the TSO. The Character Map is too large to send in a single frame and is loaded into the TSO in fragments. To improve transfer efficiency, each Character Map fragment is intentionally not digitally signed by the DCM.
6. DCM will digitally sign a frame containing the hash of the Character Map. The TSO will verify the authenticity of the preceding Character Map using the supplied hash. Once the Character Map has been verified, the TSO will set the Character Map loaded bit in the Loaded-Elements byte of the TSO status.
7. DCM will send the Pixel Interceptor configuration. Once the configuration has been verified, the TSO will set the Configuration loaded bit in the Loaded-Elements byte of the TSO status.
8. DCM will send a message overlay to the TSO for rendering to the screen. At this point the session is active, the Character Map and Pixel Interceptor configuration is valid and loaded in the TSO.

Checking signatures

Frames that are checked for signatures:

- COM_STATUS_REQ (when requested in the frame)
- COM_CONFIG
- COM_MESSAGE
- COM_CHARACTER_MAP_HASH
- COM_START_SESSION_DCM
- COM_START_SESSION_TSO
- COM_STATUS_TSO

TSO status:

State	Value
SESSION_STATE_DEFAULT	0x00
SESSION_STATE_BLANK	0x01
SESSION_STATE_NO_SESSION	0x02
SESSION_STATE_SESSION_ACTIVE	0x04

TSO errors (more then one can be set at the same time):

Error	Value
ERROR_SIGNATURE	0x01
ERROR_CNT	0x02
ERROR_SESSION_ID	0x04
ERROR_CONFIG_INCORRECT	0x08
ERROR_HASH	0x10
ERROR_TIMEOUT	0x20
ERROR_STATE_SEQUENCE	0x40
ERROR_CORRUPTED_MEMORY	0x80

TSO loaded elements (more then one can be set at the same time):

Element	Value
CHARACTER_MAP	0x01
CONFIGURATION	0x02
REMOTE_KEY	0x04
CA_KEY	0x08
HOST_KEY	0x10
EEPROM_MEMORY	0x20

Frame format

When sending 16 or 32 bit values in the Data field make sure it is Least Significant Byte First (Little-Endian).

<table>
<tr><td colspan="7">Communication frame, constant length of 512 bytes</td></tr>
<tr><td colspan="5">Data section
444 bytes</td><td>Signature section
64 bytes</td><td>CRC section
4 bytes</td></tr>
<tr><td>Command</td><td>Frame number</td><td>CNT</td><td>Session ID</td><td>Data</td><td rowspan="2"></td><td rowspan="2"></td></tr>
<tr><td>1 byte</td><td>1 byte</td><td>2 bytes</td><td>4 bytes</td><td>436 bytes</td></tr>
</table>

Message Types

#	NAME	Description	Payload description
1	COM_STATUS_REQ	From DC meter to TSO, request for status or readout of last frame. This frame does not change anything in the state machine	Frame #: x Session ID: x CNT: x D.B.0: 0 - TSO will not check the signature, use this to get TSO readout 1 - TSO will check the signature, use it to indicate that there is a communication to DC meter and TSO should not go into out of service mode

<table>
<tr><td>2</td><td>COM_STATUS_TSO</td><td>TSO status frame, signed</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-3</td><td>Interceptor status</td></tr>
<tr><td>D.B. 4-7</td><td>Microblaze status</td></tr>
<tr><td>D.B. 8</td><td>TSO status</td></tr>
<tr><td>D.B. 9</td><td>TSO errors</td></tr>
<tr><td>D.B. 10</td><td>TSO loaded elements</td></tr>
<tr><td>D.B. 11</td><td>Reserved</td></tr>
<tr><td>D.B. 12-75</td><td>Trust Anchor public key (raw)</td></tr>
<tr><td>D.B. 76-139</td><td>TSO public key (raw)</td></tr>
<tr><td>D.B. 140-203</td><td>Signature of Trust Anchor and TSO public keys (raw)</td></tr>
</table></td></tr>
<tr><td>3</td><td>COM_MESSAGE</td><td>DC meter message data</td><td><table>
<tr><td>Frame #</td><td>Starts from 0, increment until the all the data are send</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-247</td><td>Characters in the message</td></tr>
</table></td></tr>
<tr><td>4</td><td>COM_END_SESSION</td><td>End session command from DC meter</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
</table></td></tr>
<tr><td>5</td><td>COM_START_SESSION_DCM</td><td>Start new session, from DC meter</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Session ID currently signaled by TSO</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-3</td><td>DC meter random number</td></tr>
</table></td></tr>
<tr><td>6</td><td>COM_START_SESSION_TSO</td><td>Start new session, from TSO</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>0</td></tr>
<tr><td>D.B. 0-3</td><td>DC meter random number</td></tr>
</table></td></tr>
<tr><td>7</td><td>COM_PARING_REMOVE_DCM</td><td>Command to remove paired DC meter device, from DC meter to TSO, signed either by current DCM or TA</td><td>No payload</td></tr>
<tr><td>8</td><td>COM_PARING_PROMPT</td><td>From TSO signalling that there is no paired DC meter</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Pairing session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
</table></td></tr>
<tr><td>9</td><td>COM_PARING_REQ</td><td>Frame from DC meter to TSO to pair the DCM, signed by TA</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-63</td><td>TSO public key</td></tr>
<tr><td>D.B. 64-127</td><td>DCM public key</td></tr>
</table></td></tr>
</table>

<table>
<tr><td>10</td><td>COM_CONFIG</td><td>Frame with configuration information for TSO; if configuration is acceptable it triggers the change and communicates changes to interceptor</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0</td><td>Number of characters X</td></tr>
<tr><td>D.B. 1</td><td>Number of characters Y</td></tr>
<tr><td>D.B. 2-3</td><td>Overlay field size X in pix</td></tr>
<tr><td>D.B. 4-5</td><td>Overlay field size Y in pix</td></tr>
<tr><td>D.B. 6-7</td><td>Overlay field position X in pix (absolute)</td></tr>
<tr><td>D.B. 8-9</td><td>Overlay field position Y in pix (absolute)</td></tr>
<tr><td>D.B. 10-11</td><td>Character field position X in pix (relative to overlay field)</td></tr>
<tr><td>D.B. 12-13</td><td>Character field position Y in pix (relative to overlay field)</td></tr>
<tr><td>D.B. 14</td><td>Border thickness in pix</td></tr>
<tr><td>D.B. 15</td><td>Rotation:
0 - native position
1 - rotation (ISD option)</td></tr>
<tr><td>D.B. 16-39</td><td>Color map data</td></tr>
<tr><td>D.B. 40</td><td>LVDS source, 0-FPGA, 1-HMI</td></tr>
<tr><td>D.B. 41</td><td>Screen version: 0 - ISD
1 - PK350</td></tr>
</table></td></tr>
<tr><td>11</td><td>COM_CHARACTER_MAP</td><td>Frames containing character data, signature is ignored</td><td><table>
<tr><td>Frame #</td><td>Start with 0</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-248</td><td>Character map data</td></tr>
</table></td></tr>
<tr><td>12</td><td>COM_CHARACTER_MAP_HASH</td><td>Frame with hash of the whole character map data, signed</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>Current session ID</td></tr>
<tr><td>CNT</td><td>Rolling counter</td></tr>
<tr><td>D.B. 0-3</td><td>Number of bytes in character map</td></tr>
<tr><td>D.B. 4-36</td><td>Hash of character map data</td></tr>
</table></td></tr>
<tr><td>13</td><td>COM_KEY_DCM</td><td>Frame with DCM public key</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-63</td><td>Trust Anchor public key</td></tr>
<tr><td>D.B. 64-127</td><td>DCM public key</td></tr>
<tr><td>D.B. 128-191</td><td>Signature of Trust Anchor public key and DCM public key from Trust Anchor</td></tr>
</table></td></tr>
<tr><td>14</td><td>COM_TRUST_ANCHOR_REQ</td><td>Request from TA to send key for signature</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
</table></td></tr>
</table>

<table>
<tr><td>15</td><td>COM_TSO_TRUST_ANCHOR_RESPONSE</td><td>Response from TSO with its public key, unique ID and signature of those two values</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-11</td><td>Reserved</td></tr>
<tr><td>D.B. 12-75</td><td>TSO public key</td></tr>
<tr><td>D.B. 76-139</td><td>TSO unique ID</td></tr>
<tr><td>D.B. 140-204</td><td>Signature of public key and unique ID</td></tr>
</table></td></tr>
<tr><td>16</td><td>COM_TRUST_ANCHOR_RESPONSE</td><td>Response of trust anchor with signatures</td><td><table>
<tr><td>Frame #</td><td>x</td></tr>
<tr><td>Session ID</td><td>x</td></tr>
<tr><td>CNT</td><td>x</td></tr>
<tr><td>D.B. 0-63</td><td>Trust Anchor public key</td></tr>
<tr><td>D.B. 64-127</td><td>TSO public key</td></tr>
<tr><td>D.B. 128-191</td><td>Trust Anchor signature of TSO public key</td></tr>
</table></td></tr>
</table>

Display Format

Overview

The TSO controlled section of the display will be structured as a fixed character array of X by Y characters. Each fixed size character will be mapped to a pixel map array.

This will allow the DCM to have control of the information displayed, whilst minimising the load on communications.

Iconography can still be used but any icon larger than 1 character will need to be converted into 1 character sections.

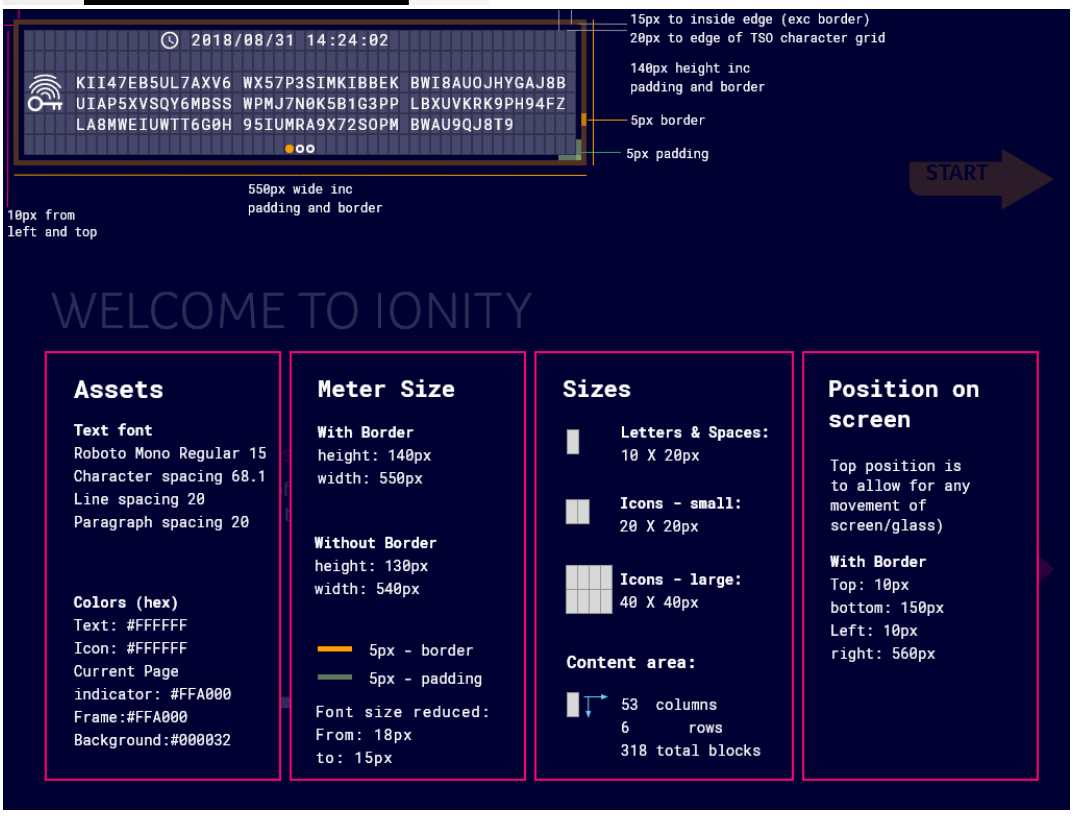
There will be eight colours loaded on startup: foreground colour (shades: 1-4), background colour, accent colour, red and green.

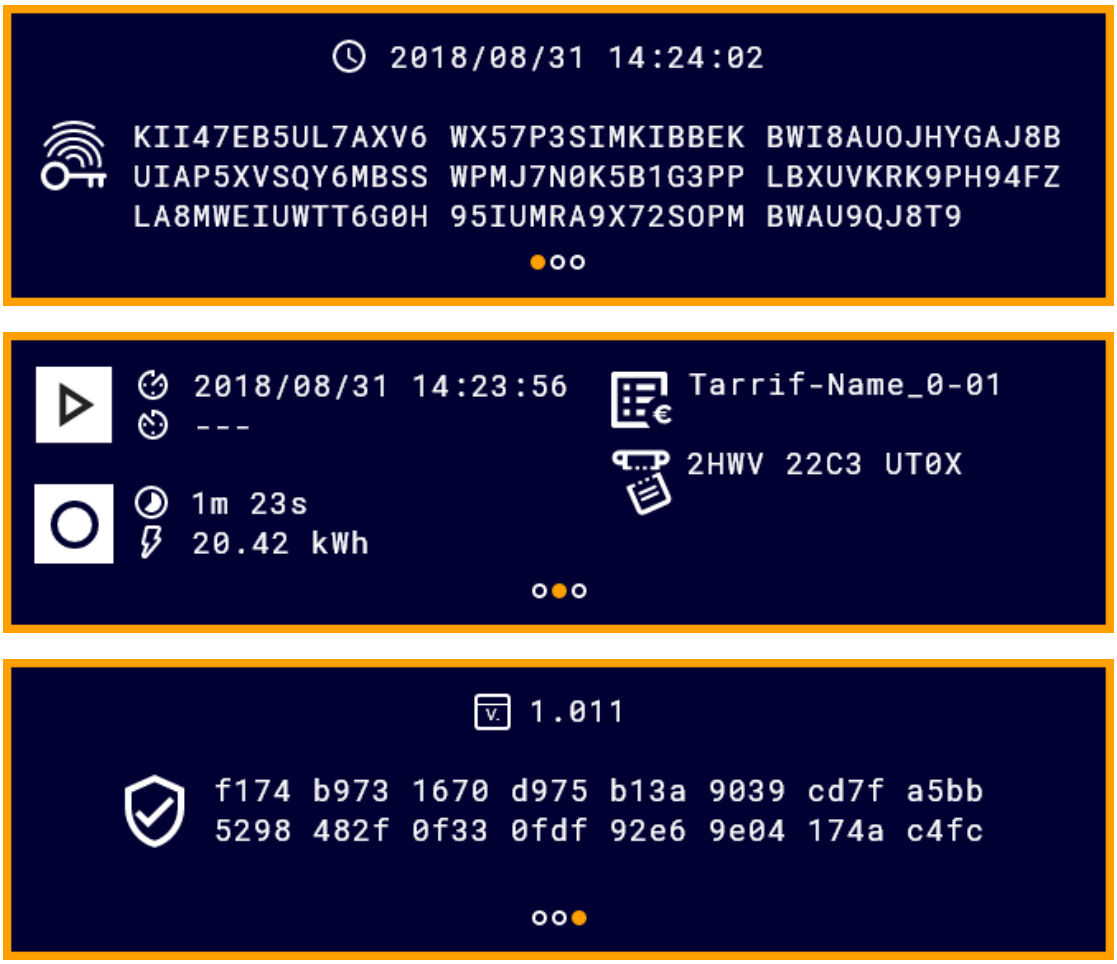
Session energy has sufficient room for more than 7 whole number digits in kWh and 3 decimal place digits.

The three screens shown are cycled through, each screen is displayed for 5 seconds.

Example:


Meter Size
With Border
height: 182px
width: 674px
Without Border
height: 167px
width: 659px
5px - border
Padding:
14px x 14px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Bottom position is to allow for any movement of screen/glass)
With Border
Top: 1091px
bottom: 1265px
Left: 67px
right: 734px
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #FFA000
Frame:#FFA000
Background:#000032
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9


2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
15px to inside edge (exc border)
20px to edge of TSO character grid
140px height inc padding and border
5px border
5px padding
550px wide inc padding and border
10px from left and top
START
WELCOME TO IONITY
Assets
Text font
Roboto Mono Regular 15
Character spacing 68.1
Line spacing 20
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #FFA000
Frame:#FFA000
Background:#000032
Meter Size
With Border
height: 140px
width: 550px
Without Border
height: 130px
width: 540px
5px - border
5px - padding
Font size reduced:
From: 18px
to: 15px
Sizes
Letters & Spaces:
10 X 20px
Icons - small:
20 X 20px
Icons - large:
40 X 40px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Top position is to allow for any movement of screen/glass)
With Border
Top: 10px
bottom: 150px
Left: 10px
right: 560px



Responsibilities

DCM:

- Non-volatile storage of pixel map of characters
- Determines characters to display on update of screen
- Manages cycling of different screens

TSO:

- Loads pixel map of characters/colours on startup
- Checks validity of configuration
- When no connection to DCM displays static error screen
- Converts character array to pixel display

Upgrade/Customisation Impact

Change Required	Resultant Change	Involves Re-Certification Involvement
Change colours	Reconfiguration	No
New location for display area	Reconfiguration	No
Change font; within fixed character size	New DCM Image - Bootloadable	Yes?
Change font; outside fixed character size	New DCM Image; New FPGA Image	Yes?
Change information shown within display area	New DCM Image - Bootloadable	Yes?

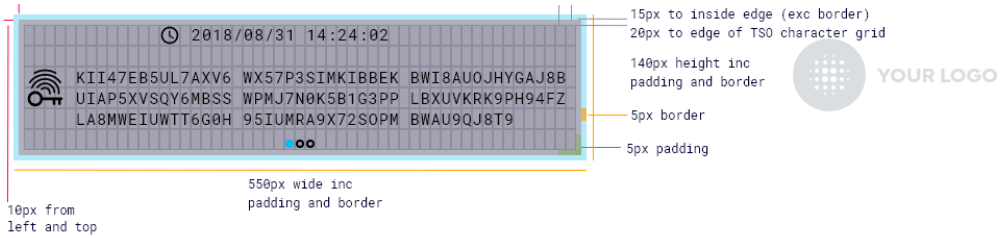
New screen (different pixel ratio/rotation/display area)	New DCM Image; New FPGA Image	Yes

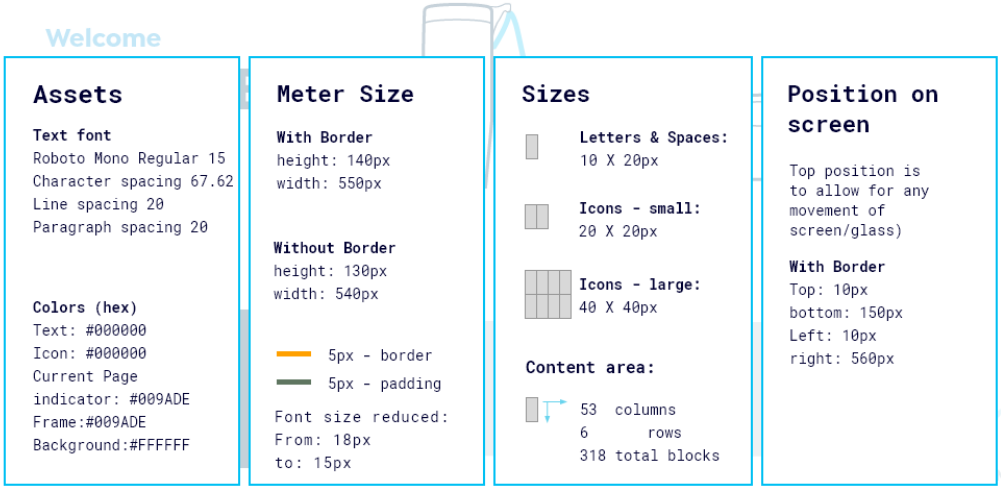
---

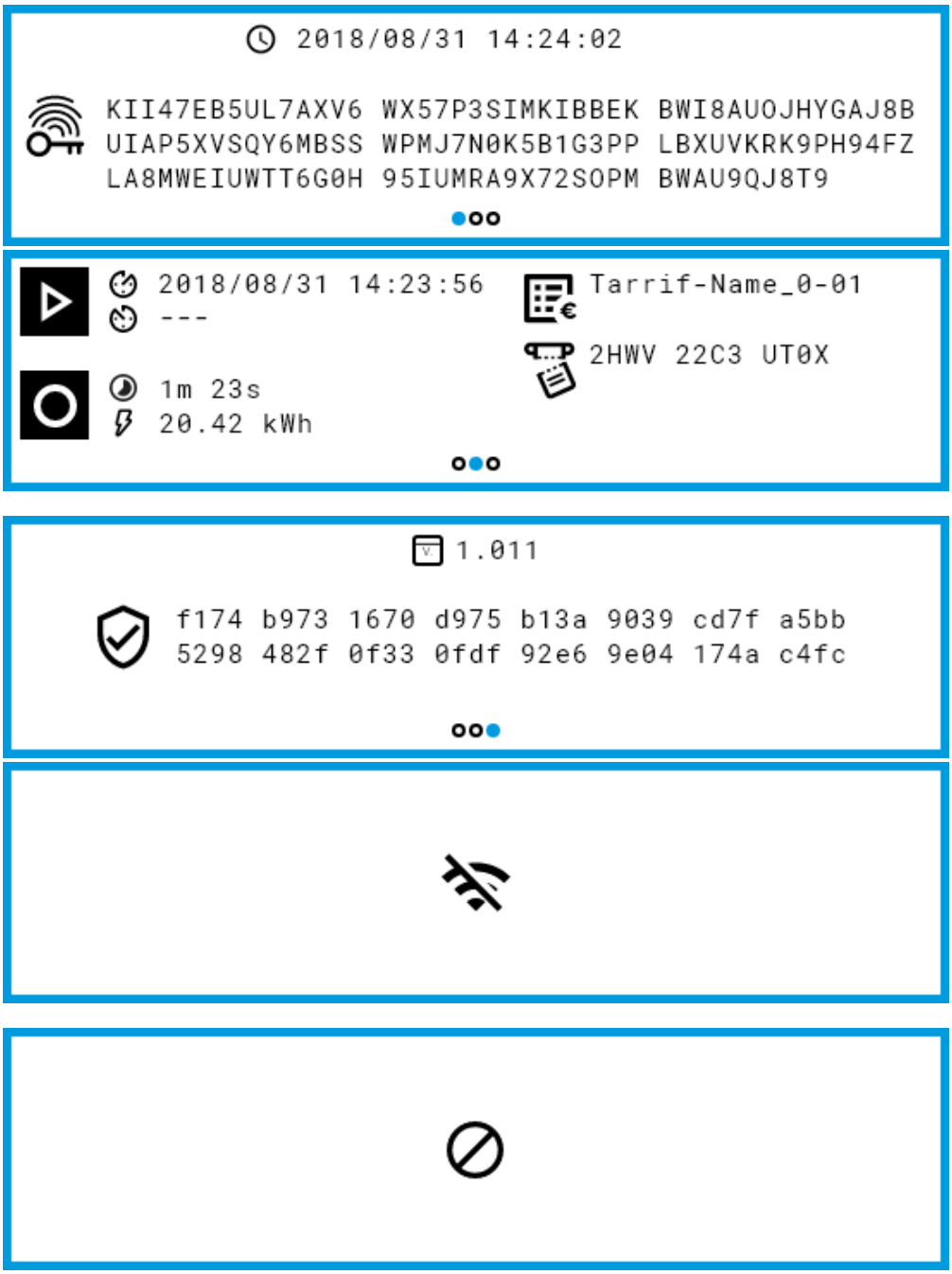
TSO Layout for Tritium's product range

Product group	Tritium PK Ionity PK	ISD	RT175s	MSC
HMI supplier	AiS	Touch Int.	Touch Int.	AiS
Resolution px WxH	1024x768	800x1280	1280x800	1280x800
Size mm WxH	305.2x229.0	136.6x217.0	217.0x136.6	217.0x136.6
Nb of TSO characters	53x6	53x6	53x6	53x6
character size px WxH	10x20	12x24	12x24	12x24
Padding px	5x5	14x14	14x14	0x0 (Tbc)
Border px	5	5	5	3
Overall size px WxH	550x140	674x182	674x182	642x150 (Tbc)

TRITIUM PK

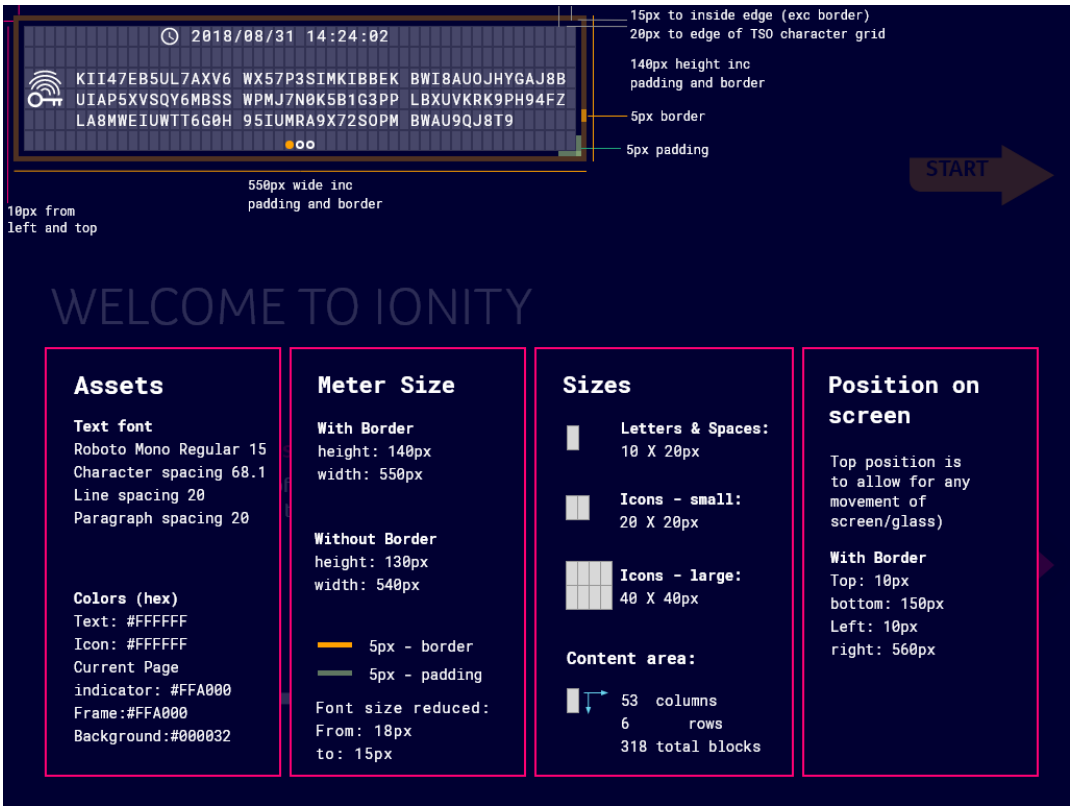


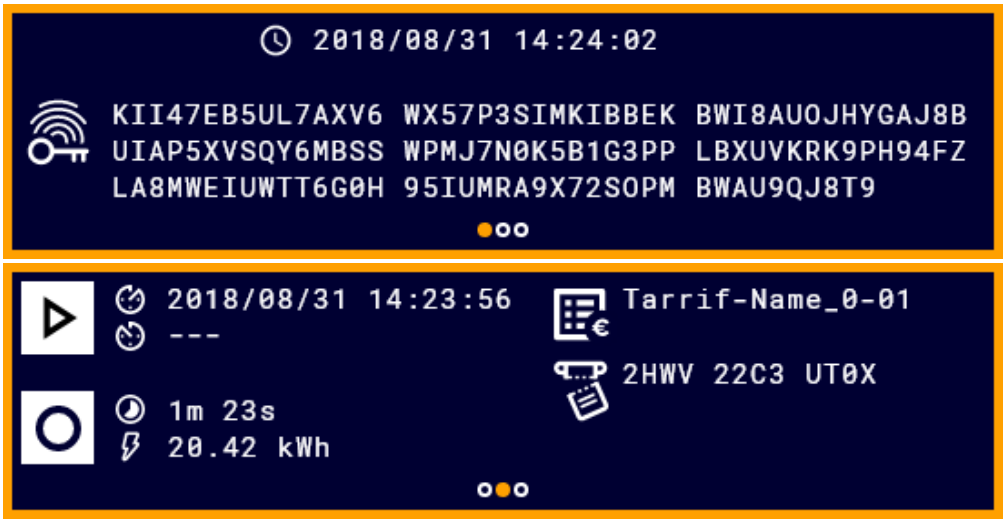



2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh
1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc

IONITY PK


2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
15px to inside edge (exc border)
20px to edge of TSO character grid
140px height inc
padding and border
5px border
5px padding
START
550px wide inc
padding and border
10px from
left and top
WELCOME TO IONITY
Assets
Text font
Roboto Mono Regular 15
Character spacing 68.1
Line spacing 20
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page
indicator: #FFA000
Frame:#FFA000
Background:#000032
Meter Size
With Border
height: 140px
width: 550px
Without Border
height: 130px
width: 540px
5px - border
5px - padding
Font size reduced:
From: 18px
to: 15px
Sizes
Letters & Spaces:
10 X 20px
Icons - small:
20 X 20px
Icons - large:
40 X 40px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Top position is to allow for any movement of screen/glass)
With Border
Top: 10px
bottom: 150px
Left: 10px
right: 560px

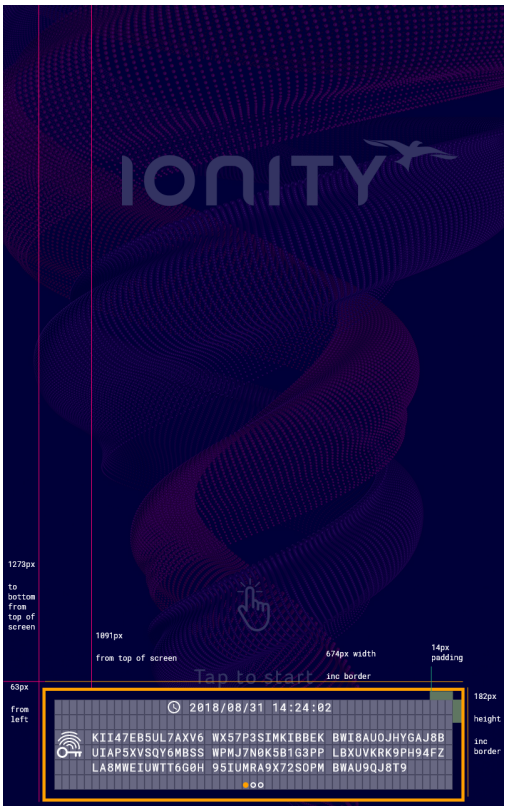

2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh


1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc

ISD


Meter Size
With Border
height: 182px
width: 674px
Without Border
height: 167px
width: 659px
5px - border
Padding:
14px x 14px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Bottom position is to allow for any movement of screen/glass)
With Border
Top: 1091px
bottom: 1265px
Left: 67px
right: 734px
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #FFA800
Frame:#FFA800
Background:#000032
Tap to start
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9

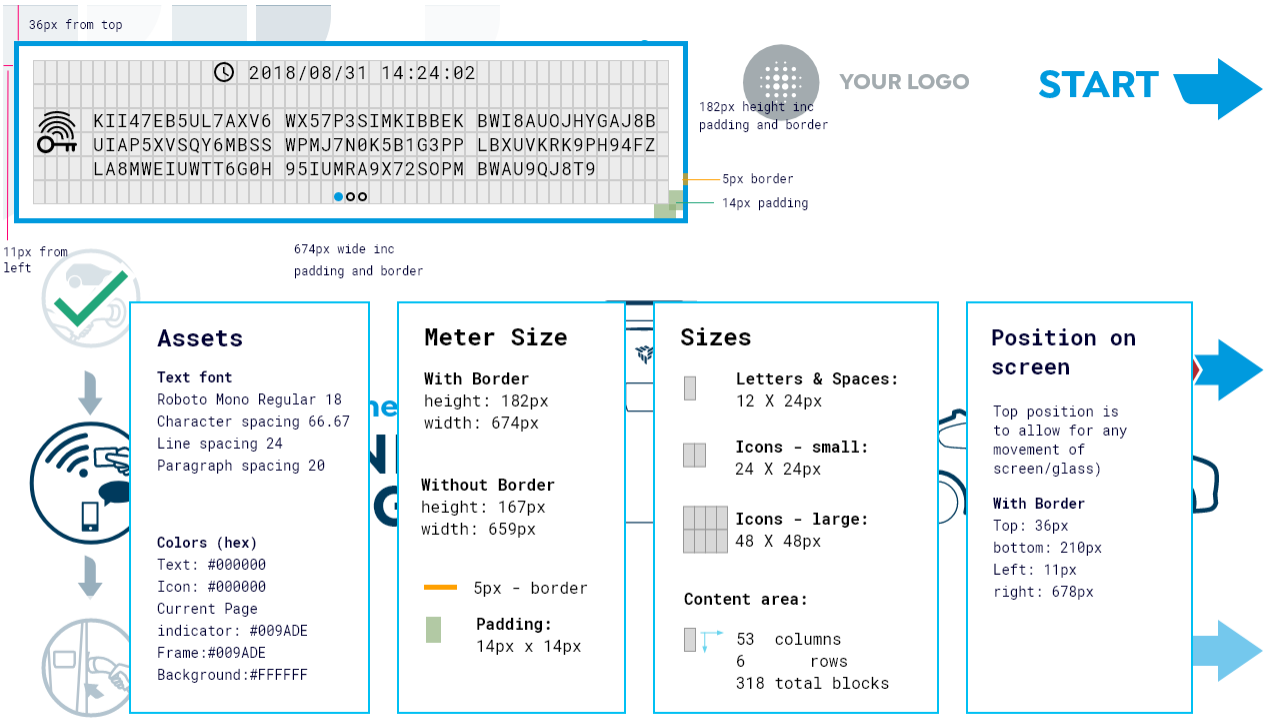

1273px
to bottom from top of screen
1091px
from top of screen
674px width
inc border
14px padding
63px
from left
Tap to start
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
182px
height
inc border

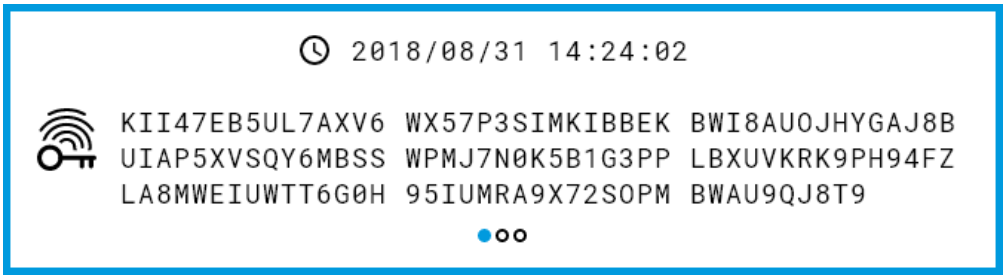

Tap to start
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9

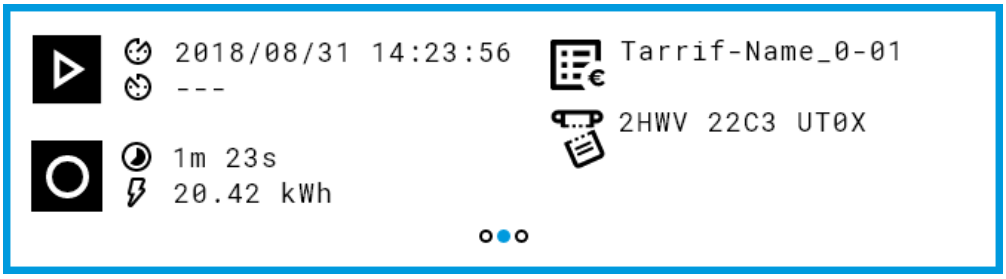

2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh
1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc

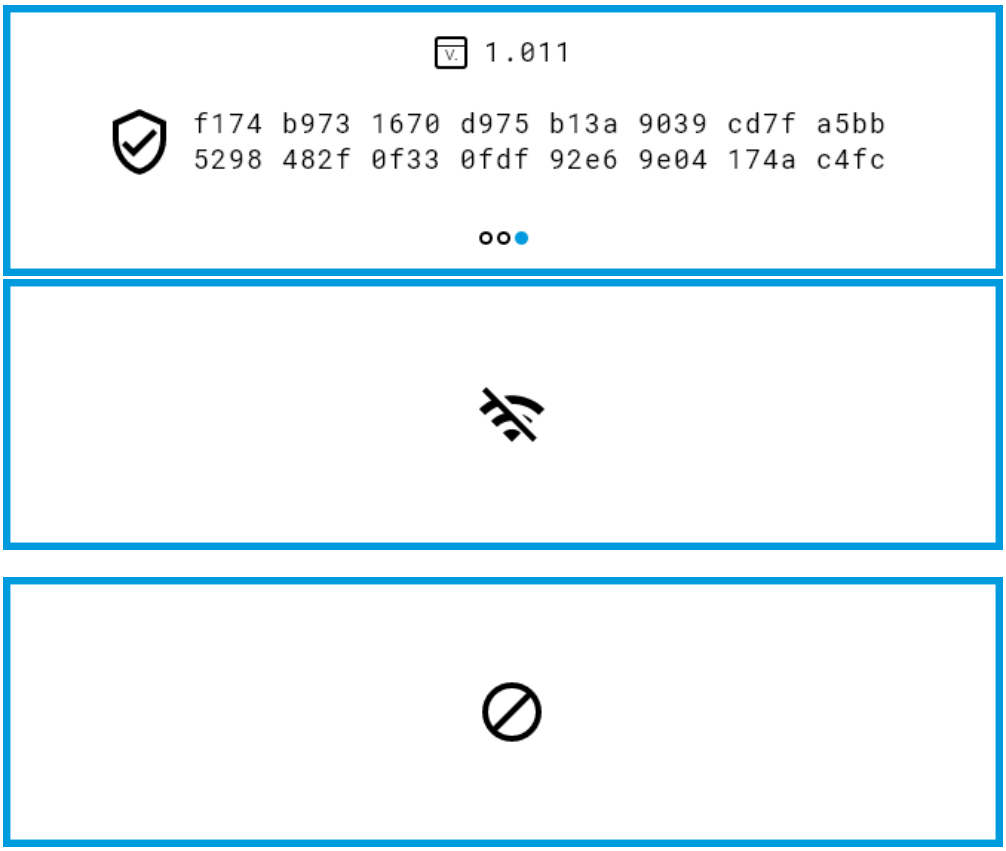
RT175s


36px from top
2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9
182px height inc padding and border
YOUR LOGO
START
5px border
14px padding
11px from left
674px wide inc padding and border
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #000000
Icon: #000000
Current Page indicator: #009ADE
Frame:#009ADE
Background:#FFFFFF
Meter Size
With Border
height: 182px
width: 674px
Without Border
height: 167px
width: 659px
5px - border
Padding:
14px x 14px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
Top position is to allow for any movement of screen/glass)
With Border
Top: 36px
bottom: 210px
Left: 11px
right: 678px


2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9

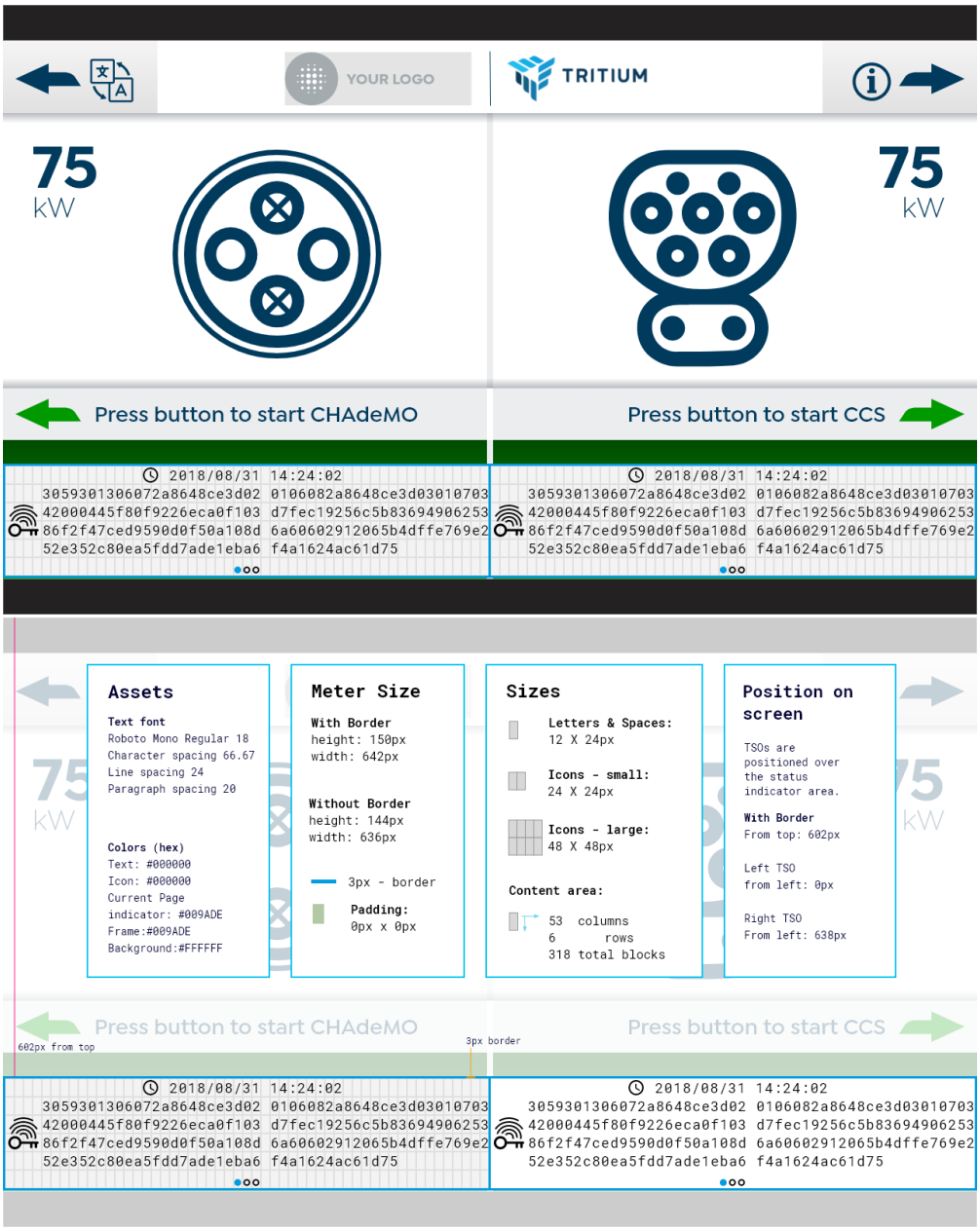

2018/08/31 14:23:56

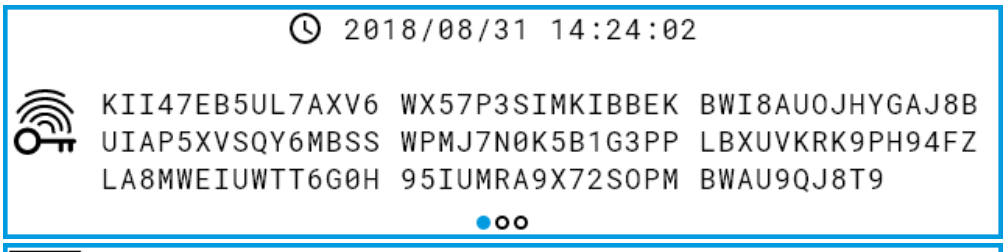
Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh



MSC (RTM/PKM)

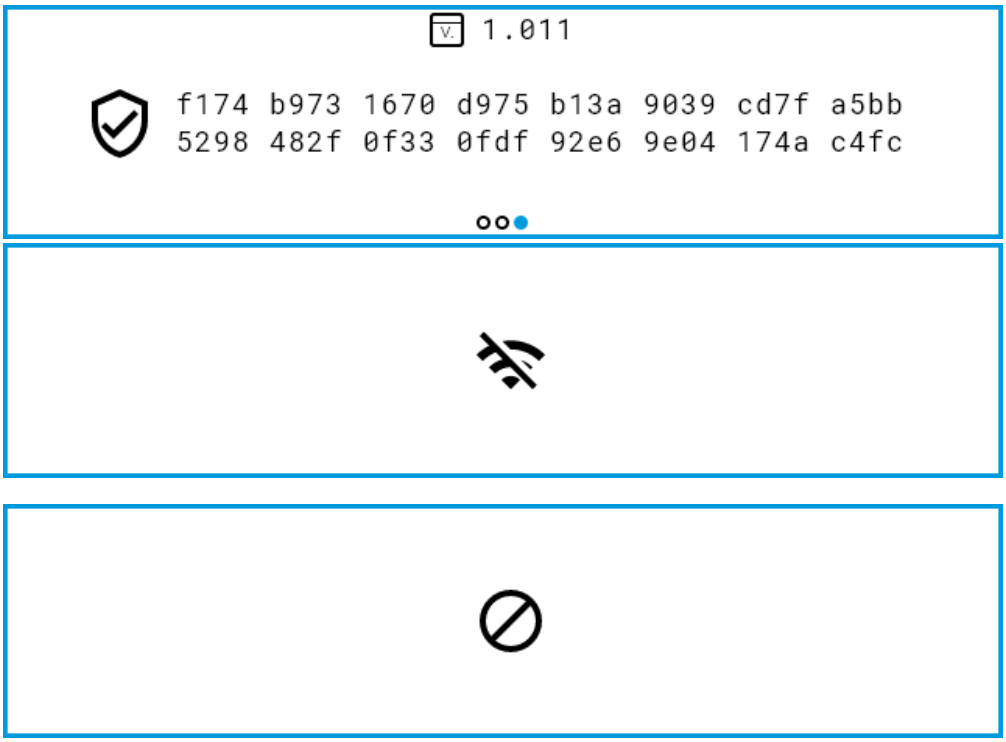
Current implementation (Nov 2021)


YOUR LOGO
TRITIUM
75
kW
75
kW
Press button to start CHAdeMO
Press button to start CCS
2018/08/31 14:24:02
3059301306072a8648ce3d02 0106082a8648ce3d03010703
42000445f80f9226eca0f103 d7fec19256c5b83694906253
86f2f47ced9590d0f50a108d 6a60602912065b4dffe769e2
52e352c80ea5fdd7ade1eba6 f4a1624ac61d75
Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #000000
Icon: #000000
Current Page indicator: #009ADE
Frame:#009ADE
Background:#FFFFFF
Meter Size
With Border
height: 150px
width: 642px
Without Border
height: 144px
width: 636px
3px - border
Padding:
0px x 0px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
TSOs are positioned over the status indicator area.
With Border
From top: 602px
Left TSO
from left: 0px
Right TSO
From left: 638px
602px from top
3px border

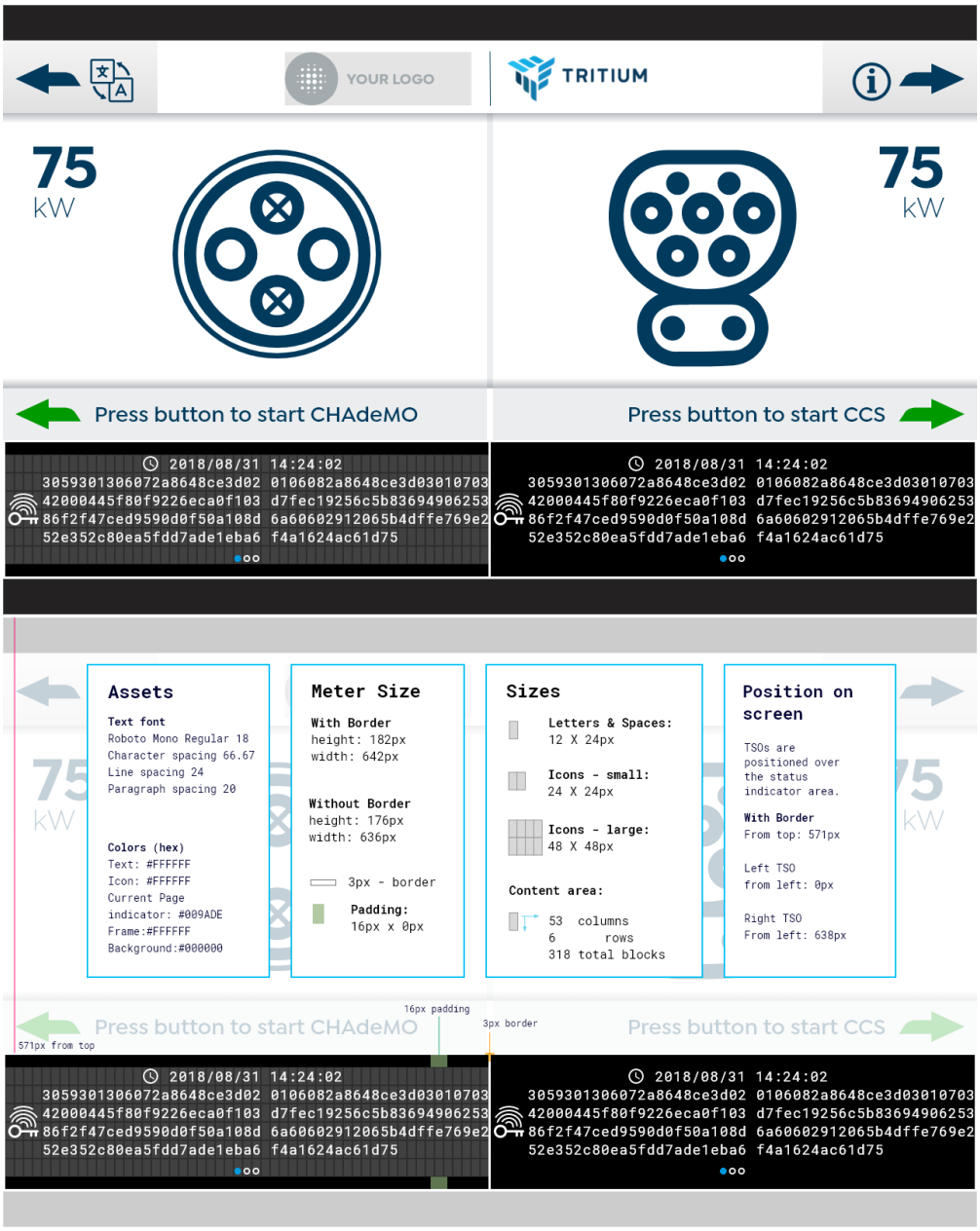

2018/08/31 14:24:02
KII47EB5UL7AXV6 WX57P3SIMKIBBEK BWI8AUOJHYGAJ8B
UIAP5XVSQY6MBSS WPMJ7N0K5B1G3PP LBXUVKRK9PH94FZ
LA8MWEIUWTT6G0H 95IUMRA9X72SOPM BWAU9QJ8T9


2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh



Alternative proposal with larger padding top and bottom to match place holder zone :


YOUR LOGO
TRITIUM
75
kW
75
kW
Press button to start CHAdeMO
Press button to start CCS
2018/08/31 14:24:02
3059301306072a8648ce3d02 0106082a8648ce3d03010703
42000445f80f9226eca0f103 d7fec19256c5b83694906253
86f2f47ced9590d0f50a108d 6a60602912065b4dffe769e2
52e352c80ea5fdd7ade1eba6 f4a1624ac61d75

Assets
Text font
Roboto Mono Regular 18
Character spacing 66.67
Line spacing 24
Paragraph spacing 20
Colors (hex)
Text: #FFFFFF
Icon: #FFFFFF
Current Page indicator: #009ADE
Frame:#FFFFFF
Background:#000000
Meter Size
With Border
height: 182px
width: 642px
Without Border
height: 176px
width: 636px
3px - border
Padding:
16px x 0px
Sizes
Letters & Spaces:
12 X 24px
Icons - small:
24 X 24px
Icons - large:
48 X 48px
Content area:
53 columns
6 rows
318 total blocks
Position on screen
TSOs are positioned over the status indicator area.
With Border
From top: 571px
Left TSO
from left: 0px
Right TSO
From left: 638px
16px padding
3px border
571px from top
Press button to start CHAdeMO
Press button to start CCS

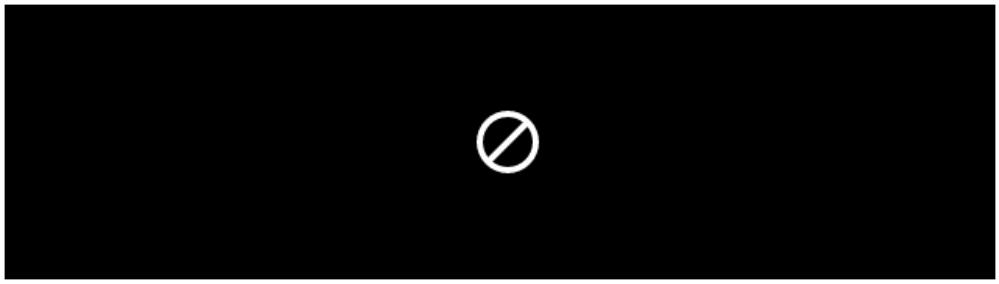

2018/08/31 14:24:02
3059301306072a8648ce3d02 0106082a8648ce3d03010703
42000445f80f9226eca0f103 d7fec19256c5b83694906253
86f2f47ced9590d0f50a108d 6a60602912065b4dffe769e2
52e352c80ea5fdd7ade1eba6 f4a1624ac61d75


2018/08/31 14:23:56

Tarrif-Name_0-01
2HWV 22C3 UT0X
1m 23s
20.42 kWh


1.011
f174 b973 1670 d975 b13a 9039 cd7f a5bb
5298 482f 0f33 0fdf 92e6 9e04 174a c4fc





FPGA IP core

Use

The Interceptor IP core can be used in a standalone VHDL or a Block Diagram project.

Functionality

- Intercept LVDS video data to LCD screen with minimum processing delay
    - Intercept logic runs at DDR, so I/OSERDES is main contributor in system delay
- Selectively override a region of the screen
- Video frame validity checking (V/H Sync, porch, visible region timing; Frame rate)
- Capable of generating LVDS video data if external video source is not present or invalid
- Switch output video source between external or internal source
    - Can be configured to be automatic or manual switching
- Clock Domain Crossing (CDC) safe memory interface to upper level of system
    - Shadow buffered frame memory (pixel colour map)
    - Shadow buffered colour map memory
    - Shadow buffered override locations
    - Interceptor control
    - Interceptor status
- Customisable display parameters
- Monitor backlight
    - Settable minimum brightness
    - Backlight fault status

Limitations

1. LVDS is presently limited to a single channel with four data lanes
2. Backlight is forced to always on
3. Override region presently limited to 122,880 pixels (Due to block RAM availability for frame buffer)
    a. This also limits the interceptor to 8 colours
    b. Depending on available memory could have DCM override area run at reduced resolution. Eg: 1 DCM pixel = 2x2 LCD pixels (4x reduction in required memory)

Interfacing

The Interceptor IP connects directly to the LVDS and backlight IO on the FPGA.

- A 100MHz input clock is required
- An active high reset signal is required

Interfacing with the upper level system can be completed:

- Directly
- Through a supporting IP core
    - Interceptor control AXI IP core to interface with an on-board CPU in the FPGA (Microblaze, ARM) (Pixel interceptor to microblaze interface)
    - *Potentially through an Interceptor control SPI IP core to interface with an external CPU from the FPGA*

To Interceptor

Sub module	Port name	Port width	Description
Frame buffer (Block RAM)	frame_addr	16	Buffer address. Valid address range: 0 - 40959 (No invalid address range checking implemented)
	frame_data	9	Buffer data. Data at each address stores a group of 3 pixels
	frame_wrEn	1	Buffer data write enable. Data is loaded at set address on rising clock clock edge
Colour map (Distributed RAM)	colour_addr	4	Buffer address. Valid address range: 0 - 7 (No invalid address range checking implemented)
	colour_data	24	Buffer data. 24 bit colour depth for each individual colour (RGB)
	colour_wrEn	1	Buffer data write enable. Data is loaded at set address on rising clock clock edge
Intercept X coordinates	ovrd_hStart	12	Start X coordinate (inclusive). Valid range: 0 - display horizontal resolution
	ovrd_hEnd	12	End X coordinate (exclusive). Valid range: 0 - displayhorizontal resolution
Intercept Y coordinates	ovrd_vStart	12	Start Y coordinate (inclusive). Valid range: 0 - display vertical resolution
	ovrd_vEnd	12	End Y coordinate (exclusive). Valid range: 0 - display vertical resolution

Interceptor general control	cpu_control	16	Control of interceptor operation
Read request strobe	read_req	1	Triggers update of 'system status' from FIFO
Backlight minimum duty cycle	bkl_duty_min	12	Backlight minimum display brightness (PWM duty cycle) NOTE: 100% duty = 'PWM period' + 1

Bitmap frame buffer data

The 3, 3bit pixel data group at each memory address of the shadow buffer is stored in the following format.

Each pixel references an element in the colour map which provides a reduced number of defined colors.

(Multi-pixel packing in a single memory address)

Bit	Description
0-2	Pixel n+2
3-5	Pixel n+1
6-8	Pixel n

NOTE: Ensure the buffer address is changed before writing new pixel data. Otherwise the pixels at the old address will be modified

Interceptor general control

The control register of the pixel interceptor operation is presented as follows

Bit	Description
0	HMI LVDS video source select
1	New frame data in shadow buffer
2	New colour data in shadow memory
3	New override X,Y location shadow values
4..15	Reserved

NOTE: The interceptor will commence loading new data at the start of the next video frame

NOTE: Bit-0 is the only signal that will work if HMI source is selected when HMI source has failed

NOTE: Bit-0 is disabled if interceptor configuration is set to AUTO change SOURCE

From Interceptor

Sub module	Port name	Port width	Description
System status	intercept_stat	32	Status of the pixel interceptor

The following status register is presented through a FIFO which the interceptor updates on the first cycle after a new video frame is started.

It is the responsibility of the upper level system to read the status register often enough such that the FIFO never becomes full.

The FIFO has an effective write depth of 63 entires.

- System Status
    - 32bit register
        - Interceptor status
            - 16bits
        - HMI frame validity status
            - 16bits

Interceptor status

Bit	Description
0	HMI LVDS video source valid
1	Frame buffer updated

2	Colour map updated
3	Override X,Y location updated
4	Frame tracking locked
5	Panel backlight fault
6	Panel backlight minimum brightness fault
7	RESERVED
8	RESERVED
9	RESERVED
10	RESERVED
11	RESERVED
12	RESERVED
13	RESERVED
14	RESERVED
15	Updated values present in this register (Excluding 'video source valid' which is always updated) NOTE: This is for the CPU to determine if the read data in this register is updated from the interceptor FIFO or from last read

Frame validity status

Bit	Description
16	hDtcErr - SyncH not detected
17	hDtcErr - SyncEn L-H not detected (Checks HBP. Also inherently VBP and VRES)
18	hDtcErr - SyncEn H-L not detected (Checks HRES)
19	hDtcErr - RESERVED
20	hCntErr - SyncH not at correct pixel location
21	hCntErr - SyncEn L-H not at correct pixel location
22	hCntErr - SyncEn H-L not at correct pixel location
23	hCntErr - HFP / Total row pixel count error (HSYNC + HPORCH[HBP] + HRES + HPORCH[HFP])
24	vDtcErr - SyncV not detected
25	vCntErr - SyncV not at correct row count
26	vCntErr - VFP / Total frame row count error (VSYNC + VPORCH[VBP] + VRES + VPORCH[VFP]) NOTE: This handles VFP detection since frame is checked against generated reference
27	RESERVED
28	RESERVED
29	RESERVED
30	vCntErr - Frame rate over error (Checked against generated reference)
31	vCntErr - Frame rate under error (Checked against generated reference)

NOTE: 'Updated' flags remain set until respective new data flag is cleared by the Microblaze

NOTE: Access to shadow buffer/memory is disabled until respective new data flag is de-asserted

NOTE: The first read of the status register will be invalid

NOTE: Bit-0 is the only signal that will work if HMI source is selected when HMI source has failed

To HW

Interface	Port name	Port width	Description
LVDS	LCD_A_CLK_P/N	1	Differential LVDS clock for Channel A to Display
	LCD_A_TX_P/N	4	Differential LVDS data for Channel Ato Display
Backlight	BKL_ON_OUT	1	Backlight enable to display backlight controller
	BKL_PWM_OUT	1	Backlight brightness PWM to display backlight controller
	BKL_FAULT_OUT	1	Backlight fault output to HMI computer
Colour mode	SEL68_BIST1	1	Display colour mode selection to display. Either 6 or 8bit colour depth
Debug	DBG_P/N	1	RESERVED
	DBG_LED	1	RESERVED

From HW

Interface	Port name	Port width	Description
LVDS	GPU_A_CLK_P/N	1	Differential LVDS clock for Channel A from HMI computer
	GPU_A_RX_P/N	4	Differential LVDS data for Channel A from HMI computer
Backlight	BKL_ON_IN	1	Backlight enable from HMI computer
	BKL_PWM_IN	1	Backlight brightness PWM from HMI computer
	BKL_FAULT_IN	1	Backlight fault output from display backlight controller

Customisation

All parameters required to use the IP core with different LCD panels can be configured through its customisation GUI.

General

Display General

Item	Description
Gen Bg colour	Internally generated frame background colour (RGB). Eg: 0xFF0000 = 100% Red, 0x00007F = 50% Blue
Use 18bit colour	Select between 18bit and 24bit colour modes (In 'general' since most screens support both)
Auto LVDS source	LVDS source is automatically changed to use internally generated signal if LVDS signal is lost

NOTE: Auto LVDS source will not switch based on frame validity. Only LVDS signal loss is used

System Clocking

Item	Description
System VCO Multiplier	Multiplier for MMCM VCO (x1000)
System Clk Divisor	System clock divisor (x1000) - System frequency must be 200MHz
Pixel Clk Divisor	Generated pixel clock divisor - Must closely match frequency of GPU pixel clock
Bit Clk Divisor	Generated bit clock divisor - Must closely match frequency of GPU bit clock
Backlight Clk PreDiv	Generated backlight clock pre-divisor - Used in conjunction with divisor
Backlight Clk Divisor	Generated backlight clock divisor - Used in conjunction with pre-divisor

Display Backlight

Item	Description

PWM period	Backlight brightness PWM period
PWM polarity	Backlight brightness PWM polarity
Fault polarity	Backlight fault polarity

PWM frequency setting

The PWM frequency is set based on the divided system clock and period configuration values.

Backlight divided clock BklClk = (100MHz × 'System VCO Multiplier') / ('Backlight Clk PreDiv' × 'Backlight Clk Divisor')

PWM period = BklClk / 'PWM frequency'

Display specific parameters

LCD General

All of these parameters must match the GPU configuration for the LCD pannel.

Item	Description
VESA colour mapping	Select between VESA and JEIDA colour mapping
Vsync polarity	LCD vertical sync active state polarity
Hsync polarity	LCD horizontal sync active state polarity
Esync polarity	LCD data enable active state polarity
V/Hsync enable	LCD requires V/H sync signals (Some LCD are 'DE only')

NOTE: GPU must always send V/H sync signals even if LCD is 'DE only' mode, due to frame validity checking logic in FPGA.

LCD Timing

All of these parameters must match the GPU configuration for the LCD pannel.

Timings can be calculated using a spreadsheet based of FPGA_G150XNE_LVDS_Calculations.xlsx

Item	Description
LVDS data rate	LVDS data rate in Mbps (in Binary Coded Decimal)
Pixel clock period	Pixel clock period (ns x1000)
Vertical resolution	LCD vertical visible area (lines)
Horizontal resolution	LCD horizontal visible area (pixels)
Vsync duration	LCD vertical sync duration (lines)
Hsync duration	LCD horizontal sync duration (pixels)
VBP duration	LCD vertical back porch duration (lines)
VFP duration	LCD vertical front porch duration (lines)
HBP duration	LCD horizontal back porch duration (pixels)
HFP duration	LCD horizontal front porch duration (pixels)
Frame rate error	Allowed horizontal line count error between selected and reference frame

NOTE: Frame rate error parameter value will depend on how closely the FPGA generated reference frame timing matches the GPU LVDS signal

Pixel interceptor to microblaze interface

The Microblaze CPU will interface with the pixel interceptor as a memory mapped AXI4-Lite peripheral which contains multiple registers than can either be written to or read from

Refer to FPGA IP core for register descriptions and usage

AXI peripheral register address mapping

The following table lists the register offset from the base address of the AXI peripheral that controls the pixel interceptor

Offset	R/W	Register
0	W	Interceptor general control
4	W	Interceptor start pixel - X
8	W	Interceptor stop pixel - X
12	W	Interceptor start pixel - Y
16	W	Interceptor stop pixel - Y
20	W	Frame buffer address
24	W	Frame buffer data
28	W	Colour buffer address
32	W	Colour buffer data
36	R	System status
40	W	Backlight minimum duty cycle

Customisation

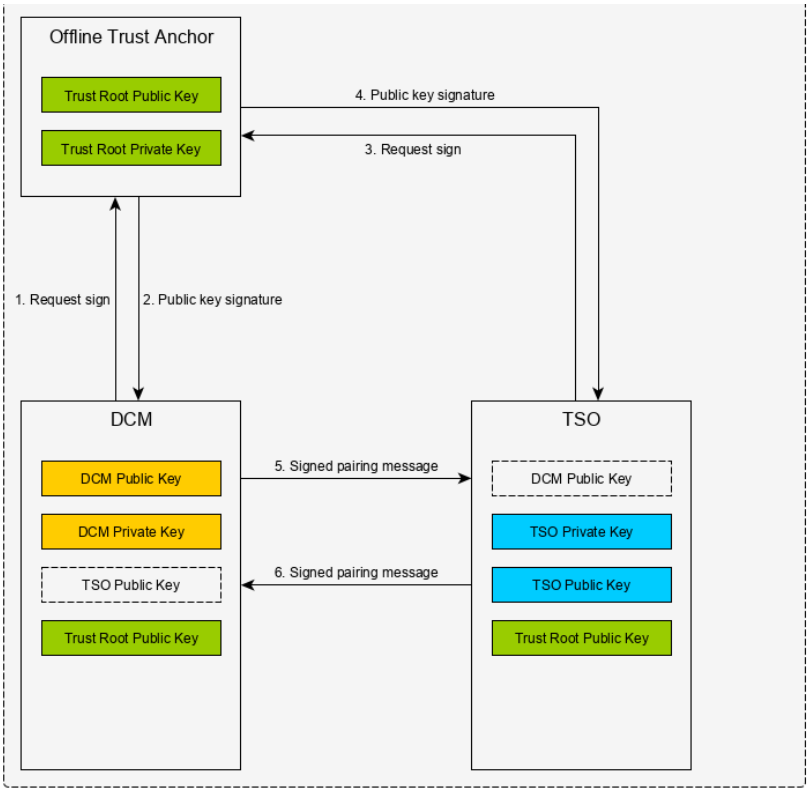
The peripheral allows for some default register values to be defined.

The default values are set after FPGA initialisation and peripheral reset.

Item	Description
Default backlight PWM min duty	Default backlight minimum brightness (PWM duty cycle) NOTE: 100% duty = 'PWM period' + 1

Pairing TSO and DCM

Secure communication is defined as either encrypted or signed.



An offline (air-gapped) trust anchor is used to sign the key of the DCM/TSO and signature is sent back to the DCM/TSO be stored. To pair a DCM and TSO together their public key and signature from trust anchor is exchanged. On verification of signature the units can determine the matching unit can be trusted and all further communication is signed using public key. The DCM and TSO save the trust anchor public key during commissioning. Trust anchor has restricted physical access and signs any device requested.

Exchange Method

- TSO is unpaired and accepts only status read frames and pairing requests. In this states it responds only with COM_PARING_PROMPT frame that contains Pairing session ID.
- DC Meter creates COM_PARING_REQ frame with pairing session ID from TSO. TSO sends its public key and the pre-saved signature.
- DC Meter sends its public key with its signature to TSO.
- DCM and TSO verify the signature with preinstalled trust root public key.
- TSO closes the pairing session and starts normal operation.

Trust Anchor Communication

Overview

The DC Meter (DCM) and Trusted Screen Overlay (TSO) need a method of proving their authenticity to each other in the field.

A simple offline Tritium-managed Trust Anchor (TA) will be used to sign public keys of DCM/TSO devices at time of production.

The public key of the TA will be loaded into all DCM/TSO devices allowing them to be able to verify signatures created by the TA at production.

Physical Interfaces

The DCM has Ethernet and is capable of TLS 1.2 communication using X.509 certificates.

The TSO only has a Serial Peripheral Interface (SPI) and doesn't support TLS 1.2 or X.509 certificates at all.

The TSO will be connected to device that provides Ethernet connectivity.

ECP Curve Type

All keys and signature will be derived from secp256r1 curve (256-bit).

Connection Type

Plain-text TCP/IP.

Port 5000 (Arbitrary unused port number).

DCM/TSO devices will be clients and initiate communication.

Message Format

Elliptic curve public key will be communicated in stripped uncompressed binary format (no prefix).

Elliptic curve public will be exactly 64 bytes.

Elliptic curve signature will be communicated in stripped uncompressed binary format (without ASN.1 encoding).

Elliptic curve signature will be exactly 64 bytes.

Client Outgoing Call

⚠ On power up, the client will generate a Elliptic curve key-pair and permanently save to non-volatile memory once TA signing is successful.

Without X.509 certificates, the Trust Anchor will not be able to verify the authenticity of the connecting client.

The authenticity will be controlled through process and physical restriction.

Version	Acknowledge	Client Public Key	Client Unique ID	Client Signature (SHA256 over Client Public Key + Client Unique ID)
1 Byte, value 0	1 Byte, value 0	64 Bytes	64 Bytes	64 Bytes

The Version value is to distinguish different message formats over time. The above format is identified by the value 0. Future formats will be identified by higher version values.

The Acknowledge value is always set to the value 0 in this request.

Client Incoming Response

Version	Acknowledge	Trust Public Key	Client Public Key	Trust Signature (SHA256 over Trust Public Key and Client Public Key)
1 Byte, value 0	1 Byte, value 0	64 Bytes	64 Bytes	64 Bytes

The Version value is to distinguish different message formats over time. The above format is identified by the value 0. Future formats will be identified by higher version values.

The Acknowledge value is always set to the value 0 in this response.

Client Outgoing Call (Acknowledgement)

Version	Acknowledge	Trust Public Key	Client Public Key	Trust Signature (SHA256 over Trust Public Key and Client Public Key)
1 Byte, value 0	1 Byte	64 Bytes	64 Bytes	64 Bytes

Acknowledge = 0, The client either did not receive or could not successfully store the information from the Trust Anchor. Trust Anchor print failure.

Acknowledge = 1, The client was able to receive and successfully store the information from the Trust Anchor. Trust Anchor print success.

Device Discoverability

The TA will not be connected to the internet.

The TA will communicate to the DCM/TSO via a network router.

The TA/DCM/TSO will communication on a IPv4 network.

The network router will provide a DHCP server.

The DCM/TSO will be capable of acquiring an IP address from a DHCP server.

The DCM/TSO assume that the TA will be at a static IP address. Alternatively the TA could supply a DNS server to provide name resolution facilities to DCM/TSO devices.

The TA will be able to support up to two concurrent connections. To potentially support up to two assemblers, one making DCM and another making TSO.

Connection Flow

1. Client start.
2. Client generates key-pair (if doesn't exist).
3. Client connects to Server using TCP socket on designated port.
4. Client uploads (Client Public Key + Unique ID + Signature).
5. Client downloads (Server Public Key + Client Public Key + Server Signature).
6. Client now has a Trust Anchor signature of it's own public key.
7. If the server encounters any issue in processing a signing request, it logs the nature of the issue and closes the TCP connection with the client.

Cryptographic Primitives and OpenSSL

The cryptographic operations and primitives described above can be emulated and verified with OpenSSL on a regular PC.

Keys

Create a key: `openssl ecparam -name prime256v1 -genkey -noout -out key.pem`

Convert to binary DER format: `openssl ec -in key.pem -outform DER -out key.der`

Extract public key: `openssl ec -in key.pem -pubout -out pub.pem`

The last 64 bytes of the file key.der correspond to the binary elliptic-curve points that go into the message format described above. They can be extracted from the key.der file, e.g., with the command `tail -c 64 key.der > key_msg_format.bin`

Signatures

To sign a message (stored in the file message_to_sign.bin) with a key (stored in the file key.pem) and store the signature in the file signature.der: `openssl dgst -sha256 -sign key.pem -out signature.der message_to_sign.bin`

The signature file contains two binary components that can be inspected with the command `openssl asn1parse -inform DER -in signature.der`:

```
  0:d=0 hl=2 l= 70 cons: SEQUENCE
  2:d=1 hl=2 l= 33 prim: INTEGER :D0423FB2F230ABD6D4F49812C6EC0D577F72E7525250D6D54FD050BD6278B986
 37:d=1 hl=2 l= 33 prim: INTEGER :AFD0A83F3AF7657957D977A15357760F1A1D625F550F525EBFEBEBD7E0CAFA5C
```

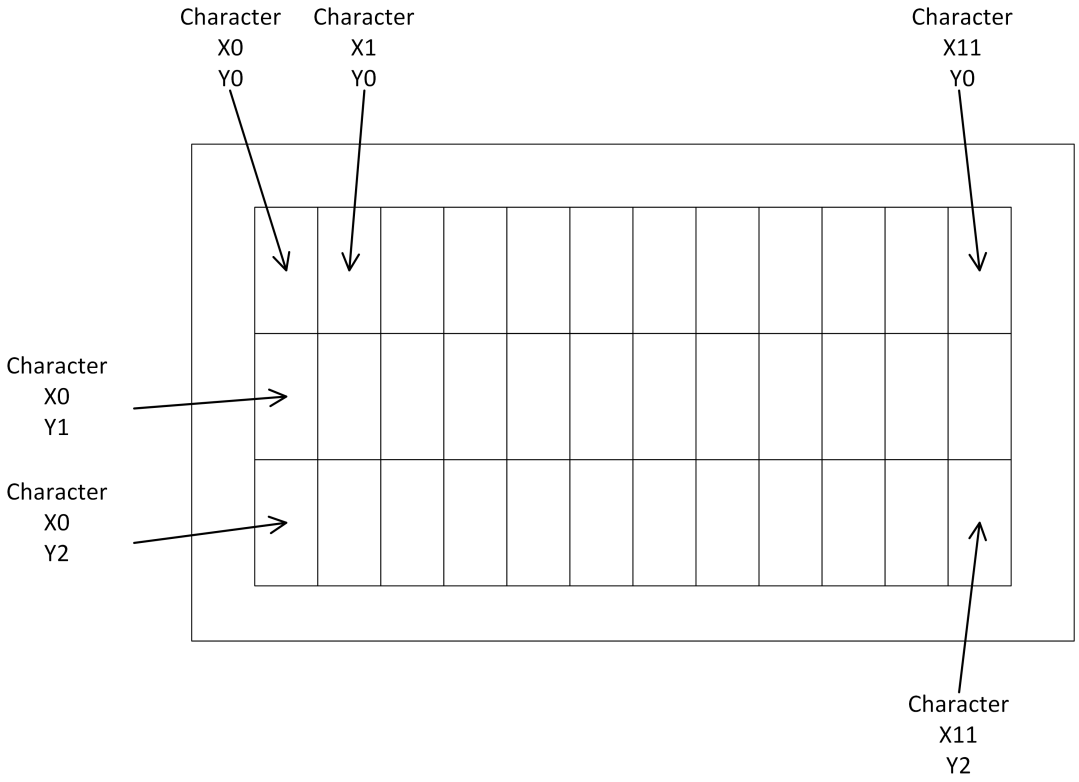
(Ignore the fact that either of them can be reported with length 33 instead of 32. This is merely an artifact of them being DER encoded. As one can see above, each hexadecimal representation is always 64 characters long, which makes for 32 bytes of binary data each.

To generate the 64-byte message format described above, the two 32-byte binary signature components simply need to be concatenated (in the order above).

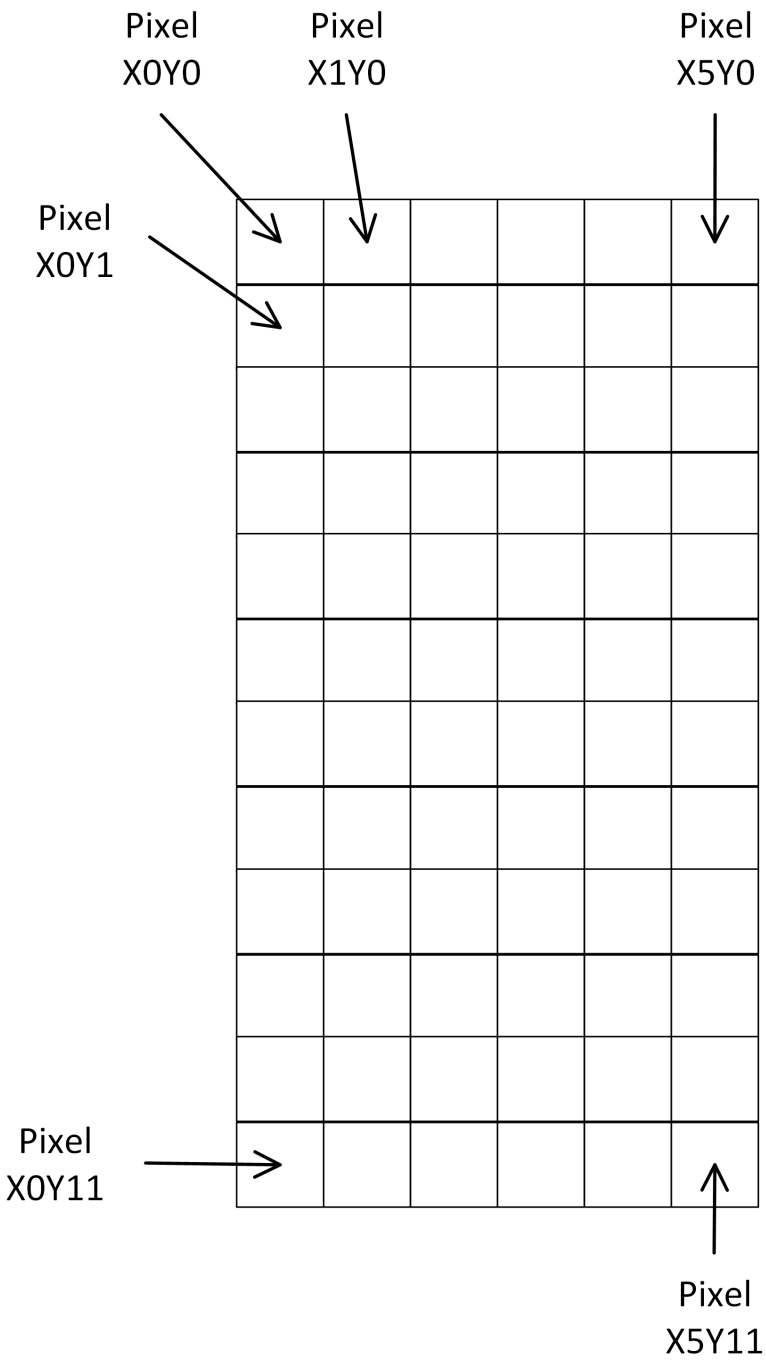
To verify a signature: `openssl dgst -sha256 -verify pub.pem -signature signature.der message_to_sign.bin`

Rendering of overlay screen

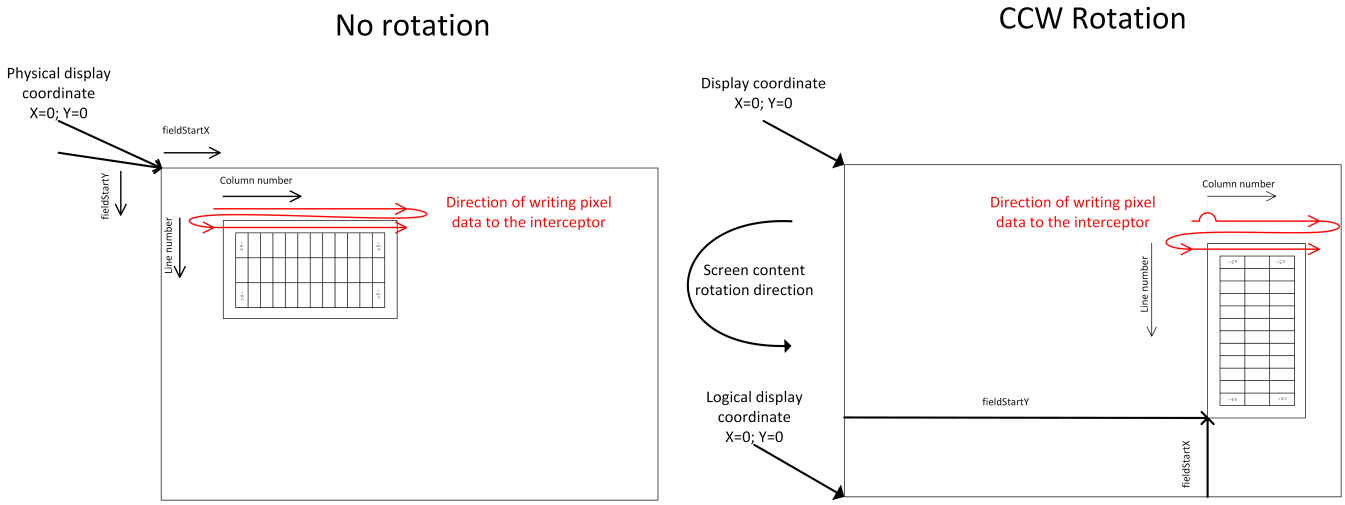
Characters in overlay field (example with character field size 12x3)



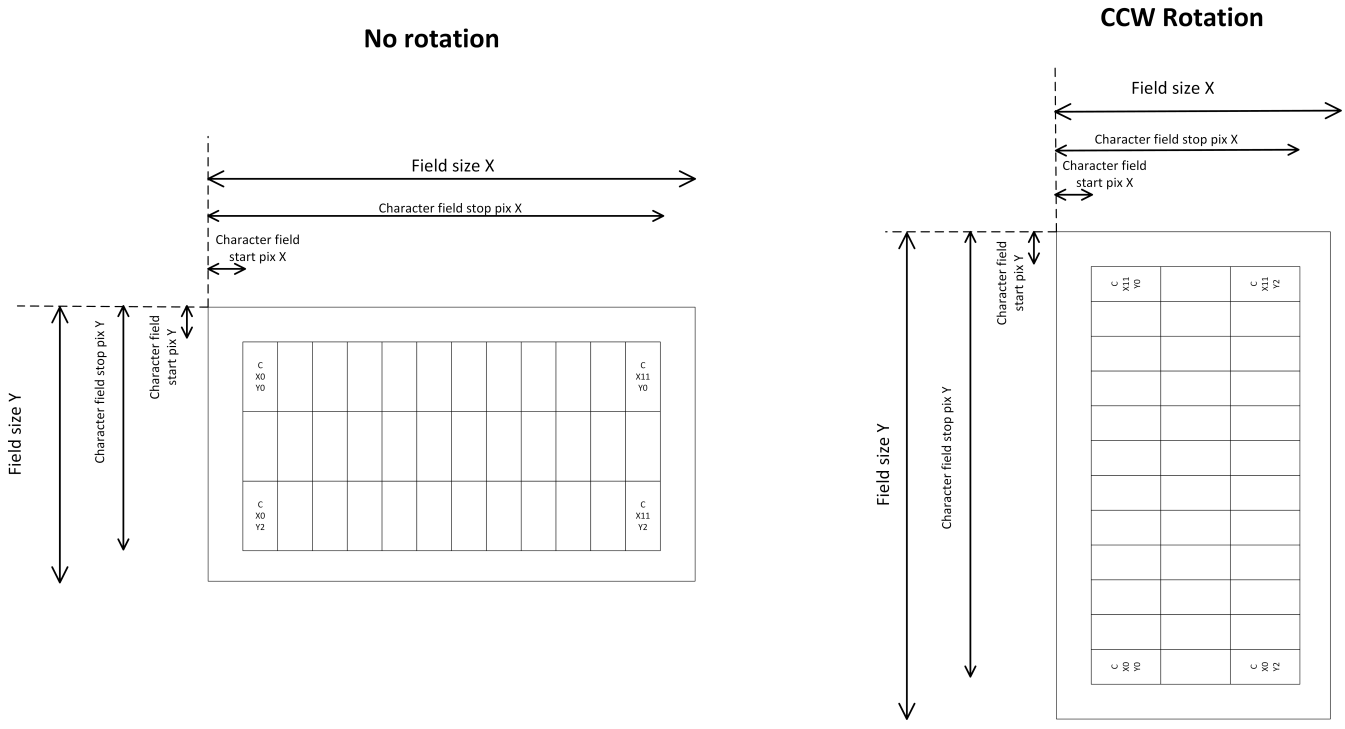
Pixels in character (example with character size 6x12)



Overlay position in normal and rotated mode



Variables used for pixel extraction (in bufferDriver_pixelExtractionContext)



DC Meter Approved Firmware Version

This page contains the firmware versions approved (or in approval) by the notified/certifying bodies.

DC Meter

Firmware version displayed in normal operation on the screen.

The application hash of the firmware is automatically generated by the firmware building pipeline and written into the image. The version number is manually controlled and incremented on release.

Status	Date Approved	Version Purpose Summary	Notified /Certification Body	Standard	Market	Version	Application Hash
In assessment	-	Original submission - overall certification	VDE	EN50470-1/-3	Germany	1.011	`f174b9731670d975b13a9039cd7fa5bb5298482f0f330fdf92e69e04174ac4fc`
Approved	TBD	PK Chargers	VDE	EN50470-1/-3	Germany	1.010	`a0f41f2ebf3ce9cdcb0833f75649627b3484321f40e422302cba12cdca565c0c`

TSO

Firmware version displayed on start-up on the screen.

The version number is manually controlled and incremented on release (more details see Trusted Screen Overlay (TSO)).

Status	Date Approved	Version Purpose Summary	Notified/Certification Body	Standard	Market	Approved Firmware Version (s)
In assessment	-	Original submission - overall certification	VDE	EN50470-1/-3	Germany	01.00.01_PK 01.00.01_ISD

Charger power derate on TCU

Overview

In PK and ISD chargers, Thermal Control Unit (TCU) managers the internal cooling system. The primary goal for internal liquid cooling is good thermal management to maintain performance and reliability.

The functionality of TCU is to:

- monitor the temperature using thermal sensors
- control fans(heat ex-changer)
- passive circulation of liquid coolant through the pumps
- heat dissipation of the coolant through radiators

Power Derate

When ambient temp is high or cooling component malfunctions, the thermal gain during high current charge may exceed the cooling capacity, in such cases TCU requests Charge Control Unit (CCU) CCU to derate the power delivered to vehicle. Otherwise, high temperature inside the charger may damage certain components, especially the power electronic compoents. The power derate is calculaed based on the temperature of internal air and liquid coolant.

$derate_{TCU}$ is based on two separate calculation: $derate_{air}$ and $derate_{coolant}$.

$derate_{TCU} = min(derate_{air} , derate_{coolant})$

$derate_{air}$ is calculated based on internal Air temperature. The derate starts from 70°C (100%) and reaches full derate 5% at 85°C, as illustrated in Figure 1. The slope of the line is 6.667.

$derate_{air}(\%) = 100 – ((temp - 70) * 6.667)$

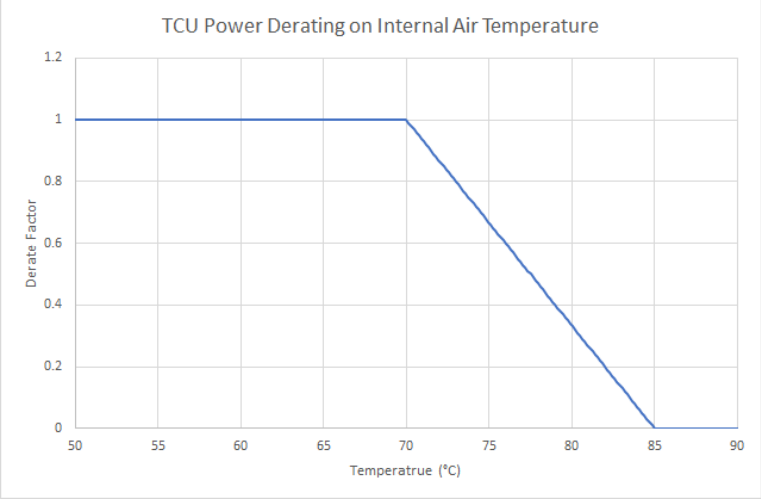


Figure 1 TCU power derating on Internal Air Temperature

$derate_{coolant}$ is calculated based on Coolant temperature. The derate starts from 65°C (100%) and reaches full derate 5% at 70°C, as illustrated in Figure 2. The slope of the line is 20.

$derate_{coolant}(\%)= 100 – ((temp - 65) * 20)$

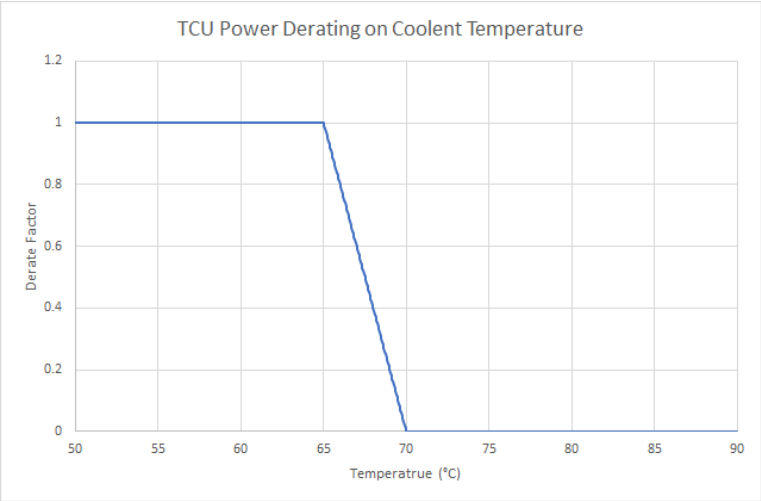


Figure 2 TCU power derating on Coolant Temperature